

Bangladesh University of Professionals



Faculty of Science and Technology Department of Information & Communication Technology (ICT)

Course Title: Database Management System Laboratory

Course Code: ICE-2204

Project Report

Project Name: Exoplanetary Management System

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Section-A, BICE-2024

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Phase-01:

Introduction:

The discovery of exoplanets has emerged as one of the most transformative advancements in modern astronomy, reshaping our understanding of planetary formation, habitability, and the potential for life beyond Earth. With thousands of exoplanets documented through space observatories and ground-based missions, researchers require reliable systems to store, analyze, and interpret large volumes of planetary data. However, traditional data repositories are often scattered, manually maintained, or lack interactive analytical tools, making it difficult for students, astronomers, and researchers to explore exoplanet information efficiently.

ExoDiscover – Exoplanet Exploration and Data Management System addresses this gap by integrating a structured Database Management System (DBMS) with an interactive web-based interface. Designed as a centralized digital hub, ExoDiscover stores comprehensive information about planets, stars, discoveries, research papers, and observational methods. The system uses MySQL for backend data management and Java Servlet + JSP for the web application, ensuring secure authentication, smooth data retrieval, and fast processing of analytical queries.

The primary purpose of ExoDiscover is to streamline the entire workflow of exoplanet exploration: managing planetary data, visualizing discovery trends, relating planets to their host stars, and understanding system characteristics more intuitively. The database architecture consists of interconnected entities including Planet, Star, Planetary System, Discovery Method, Facility, Research Paper, User, and AuditLog. These tables are linked following relational principles that ensure data consistency, accuracy, and referential integrity.

ExoDiscover supports three categories of users—Guests, Registered Users, and Administrators—each receiving role-based access to features.

Guest users can browse planets but cannot access research papers or download files, ensuring controlled access.

Registered users can explore planets more deeply, view analytical insights, and interact with discovery trends.

Administrators can upload papers, manage star and planet data, review system logs, and track activity through advanced analytics.

One of the unique strengths of ExoDiscover is its use of complex analytical SQL queries and interactive charts. Features such as hottest planets list, most packed star systems, heaviest planetary systems, discovery trends per year, paper upload frequency, and user activity logs demonstrate advanced DBMS capabilities, combining multi-table joins, grouping, filtering, aggregation, and visualization.

Summary:

ExoDiscover is an intelligent exoplanet exploration and data-management system developed using Java Servlets, JSP, and MySQL DBMS. It provides an interactive platform for browsing, analyzing, and understanding exoplanets, stars, and planetary systems stored in a relational database. The system maintains well-structured scientific entities such as Planet, Star, Planetary System, Discovery Method, Research Paper, User, and Audit Log, ensuring organized and reliable astronomical data management.

The platform supports multiple user roles—Guest, User, and Admin. Guests can only view basic planet information, whereas registered users can perform searches, explore planet details, and view scientific insights. Administrators have extended capabilities, including managing planets, stars, and research papers, and tracking all activities through a detailed audit logging system.

ExoDiscover integrates analytical features such as Top 5 hottest planets, Top 5 most packed star systems, heaviest planetary systems (by average planet mass), and discovery-trend visualizations using Chart.js. The database supports complex SQL operations involving multi-table joins, filtering, ranking, grouping, and statistical summaries to deliver these insights.

Every major action—logins, uploads, updates, deletions—is automatically recorded in the auditlog table, ensuring transparency, traceability, and accountability. The system also allows administrators to review daily paper uploads, recent activities, and user behavior patterns.

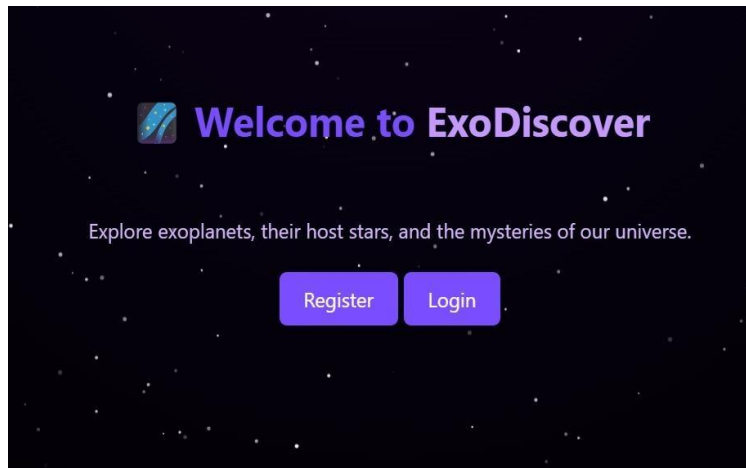
By transforming raw astronomical data into a structured, searchable, and visually engaging platform, ExoDiscover demonstrates the power of database-driven applications. It showcases strong DBMS principles, efficient data retrieval, role-based access control, and interactive scientific exploration suited for both academic and research environments.

Functions & Operations

Below are the primary functions along with their descriptions that will be included in the project.

1. Home Page:

Discription: The **ExoDiscover** home page introduces the system with a clean, user-friendly design that invites users to explore exoplanetary data. It provides direct entry points for registration and login, enabling secure access to the database functionalities.



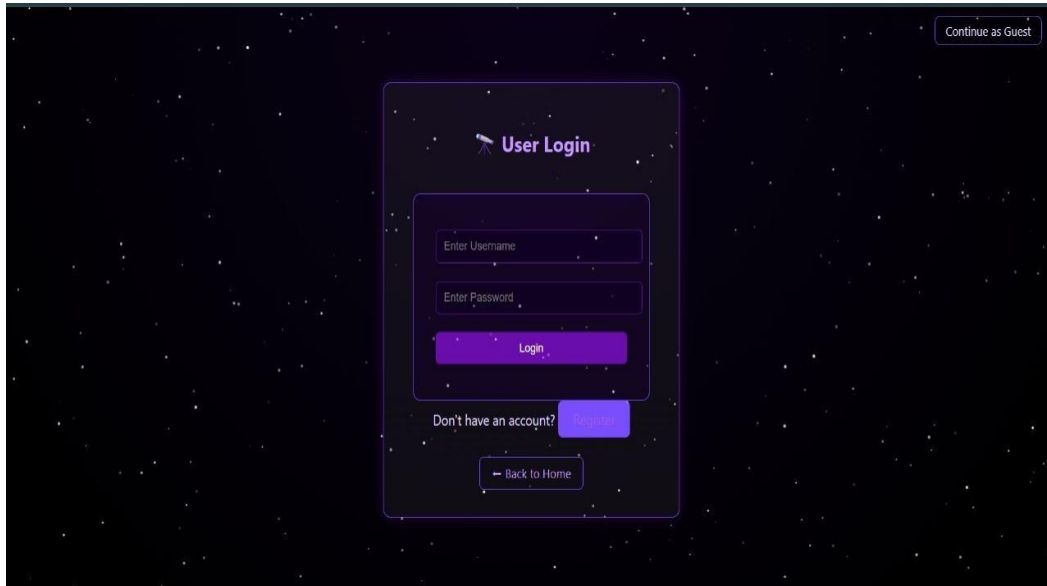
2. Registration:

Description: The *Create Account* page allows new users to register by providing a username, email address, and password. The form includes validation fields such as password confirmation to ensure accurate user input. After completing the form, users can sign up to gain access to the ExoDiscover system. A "Back to Home" button provides easy navigation, and an option is available for returning users to switch to the sign-in page.

A screenshot of the "Create Account" form. The form is centered on a dark space background with stars. It has a title "Create Account" at the top. The form contains four input fields: "Username" with the value "Lira", "Email" with the value "lira123@gmail.com", "Password" with masked characters "*****", and "Confirm Password" with masked characters "*****". Below the input fields are two buttons: "Sign Up" and "Back to Home". At the bottom of the form, there is a link "Already have an account?" followed by a "Sign in" button.

3. User login Page/Sign in:

Description: The User Login page allows registered users to securely access the system by entering their username and password. It also provides options to navigate back to the home page, create a new account, or continue as a guest.



The screenshot shows a 'User Login' page with a dark space-themed background. At the top right is a 'Continue as Guest' button. The main login form is centered and contains fields for 'Enter Username' and 'Enter Password', followed by a 'Login' button. Below the form, there is a link 'Don't have an account? Register' and a 'Back to Home' button.

4. Continue as Guest:

Description: This page allows guest users to explore the exoplanet database without creating an account. It provides a search interface where users can filter exoplanets by name, year range, discovery method, and facility, and displays matching results with options to view detailed information.



The screenshot shows the 'Explore Exoplanets' page. It features a search bar with filters for 'Name' (containing '55 cnc'), 'Year' (containing '2025'), 'Year to', 'Method', and 'Facility'. A 'Search' button is on the right. Below the search bar is a table of search results.

Planet	Host Star	Year	Method	Facility	Distance (pc)	Period (days)	Details
55 Cnc B b	55 Cnc B	2025	Radial Velocity	Mauna Kea Observatory	12.4773	6.7994	View
55 Cnc B c	55 Cnc B	2025	Radial Velocity	Mauna Kea Observatory	12.4773	33.747	View

ExoDiscover
Back to Search

55 Cnc B b

Host star: 55 Cnc B

Research Papers

Please log in to view research papers.

Login

Overview



55 Cnc B b was discovered in 2025 using the Radial Velocity method at Mauna Kea Observatory. It orbits the star **55 Cnc B** with an orbital period of **6.8 days** at a semi-major axis of **0.04 AU** and orbital eccentricity **1.69**. Physically, **55 Cnc B b** has a mass of **0.00 Earths**, a radius of **3.50 Earth radii**. The system is approximately **12.5 parsecs** from Earth. The host star is **M4.5 V**, T_{eff} **3286 K**, radius **0.27 R_☉**, mass **0.26 M_☉**, metallicity **0.16 dex**, log g **4.86**. Sky position: RA **133.17°**, Dec **28.32°**.

Key Facts

Planet	55 Cnc B b
Host Star	55 Cnc B
Discovery Year	2025
Discovery Method	Radial Velocity
Facility	Mauna Kea Observatory
Distance (pc)	12.4773
Orbital Period (days)	6.7994

5. Admin Dashboard:

Description: The Admin Dashboard provides centralized control over the ExoDiscover system. From here, the administrator can upload research papers, manage planet records, manage star information, and view system activity logs. The dashboard enables efficient oversight of database content and overall platform operations.

Home
Search



Welcome, Admin Lira

Manage the ExoDiscover database and research content.

Upload Research Paper
Manage Planets
Manage Stars
View Logs

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6. Upload Research Paper Page:

Description: The **Upload Research Paper** page is designed to allow users to submit, store, view, and manage research papers related to different planets or exoplanets. It has two main sections:

New Paper Upload Section :

This section allows the user to upload a new research paper by filling out a form. The following inputs are available:

- a. Planet Selection: A dropdown menu where the user selects the planet/exoplanet that the research paper is associated with.
- b. Title: A text field to enter the title of the research paper.
- c. Authors: A text field where the user lists the authors of the paper.
- d. Year: A field to input the publication year of the paper.
- e. PDF File Upload: A file input box that lets the user upload the PDF version of the research paper.
- f. Short Abstract / Notes: A text area where the user can write an abstract, summary, or important notes about the paper.
- g. Upload Button: When clicked, it uploads the paper along with its metadata to the backend or database.

Upload Research Paper					
New Paper			Uploaded Papers		
ID	Title	Planet	Year	Link	Action
7	kjhgf	11 UMi b	2012	Download	Delete
6	A secondary atmosphere on the rocky exoplanet 55 Cnc e	55 Cnc b	2024	Download	Delete
5	JWST reveals the rapid and strong day-side variability of 55 Cnc e	55 Cnc b	2024	Download	Delete
4	Characterizing planetary systems with SPIRou: questions about the magnetic cycle of 55 Cnc A and two new planets around B+	55 Cnc B b	2025	Download	Delete

Uploaded Papers Section :

This section displays a table of all previously uploaded research papers.

The table includes:

- a. ID: A unique identifier for each uploaded paper.
- b. Title: The title of the research paper.
- c. Planet : The planet or exoplanet associated with the paper.
- d. Year : The publication year.
- e. Link : A “Download” link that allows users to download the uploaded PDF.
- f. Action – Delete: A “Delete” button that allows users to remove a specific paper from the system.

ID	Title	Planet	Year	Link	Action
7	kjhgf	11 UMi b	2012	Download	Delete
6	A secondary atmosphere on the rocky exoplanet 55 Cnc i e	55 Cnc b	2024	Download	Delete
5	JWST reveals the rapid and strong day-side variability of 55 Cnc i e	55 Cnc b	2024	Download	Delete
4	Characterizing planetary systems with SPIRou: questions about the magnetic cycle of 55 Cnc A and two new planets around B+	55 Cnc B b	2025	Download	Delete

The New paper page allows Admin to upload new research papers by entering details such as planet, title, authors, year, PDF file, and notes. It then saves the paper along with its metadata to the system for future access.

New Paper

Planet
4 UMa b (star: 4 UMa) ▼

Title
Retrograde-rotating Exoplanets Experience Obliquity Excitati

Authors
Steven M. Kreyche1 , Jason W. Barnes1

Year
2021

PDF File
Choose File Kreyche_20....J_1_8.pdf

Short Abstract / Notes

Upload Paper

It also displays a list of previously uploaded papers, where users can download each file or delete entries from the system.

Paper "Retrograde-rotating Exoplanets Experience Obliquity Excitations in an Eccentricity-enabled Resonance" uploaded successfully.

New Paper

Planet
- Select planet - ▼

Title

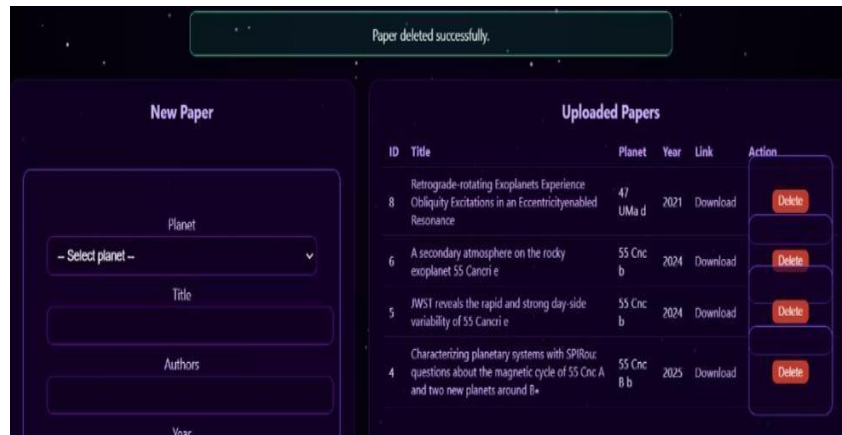
Authors

Year

Uploaded Papers

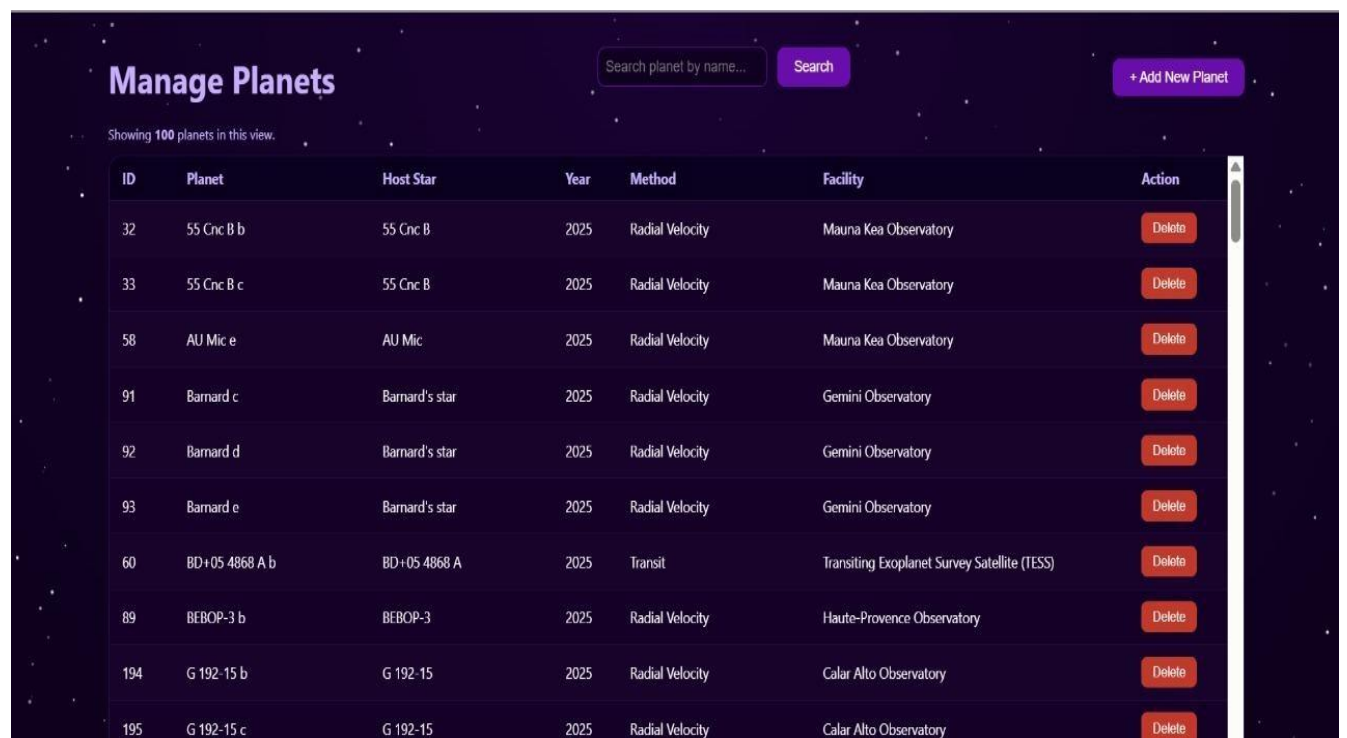
ID	Title	Planet	Year	Link	Action
8	Retrograde-rotating Exoplanets Experience Obliquity Excitations in an Eccentricity-enabled Resonance	47 UMa d	2021	Download	Delete
7	kjhgf	11 UMi b	2012	Download	Delete
6	A secondary atmosphere on the rocky exoplanet 55 Cnc e	55 Cnc b	2024	Download	Delete
5	JWST reveals the rapid and strong day-side variability of 55 Cnc e	55 Cnc b	2024	Download	Delete
4	Characterizing planetary systems with SPiRou: questions about the magnetic cycle of 55 Cnc A and two new planets around B+	55 Cnc B b	2025	Download	Delete

The Delete function allows Admin to remove an uploaded research paper from the system. When the Delete button is clicked for a specific paper, that entry—including its metadata and associated PDF—is permanently removed, and the page updates to show a confirmation message along with the revised list of remaining papers.



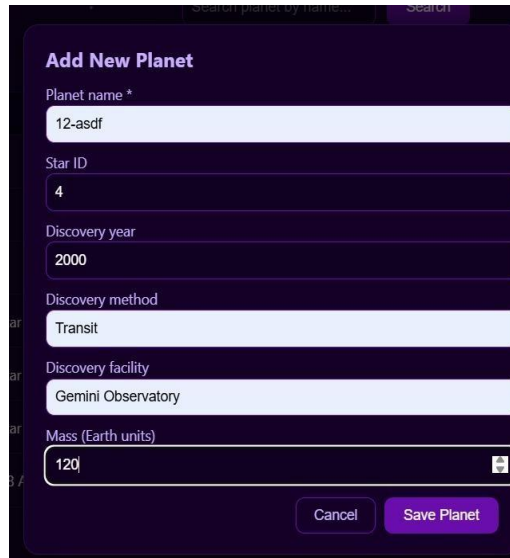
7. Manage Planets:

Description: This page allows admin to manage a database of planets by displaying them in a searchable and scrollable table. Users can view key details for each planet, including its host star, discovery year, detection method, and facility. A search bar at the top enables quick filtering by planet name, while the “Add New Planet” button lets users insert new entries into the system. Each listed planet also includes a “Delete” button, allowing users to remove specific planets from the database.



8. Add New Planet:

Description: The Add New Planet page allows admin to create a new planet entry by filling out details such as planet name, associated star ID, discovery year, method, facility, and mass. After entering the required information, users can save the planet to the system or cancel the process.



The 'Add New Planet' form is a dark-themed modal with the following fields and values:

- Planet name *: 12-asdf
- Star ID: 4
- Discovery year: 2000
- Discovery method: Transit
- Discovery facility: Gemini Observatory
- Mass (Earth units): 120

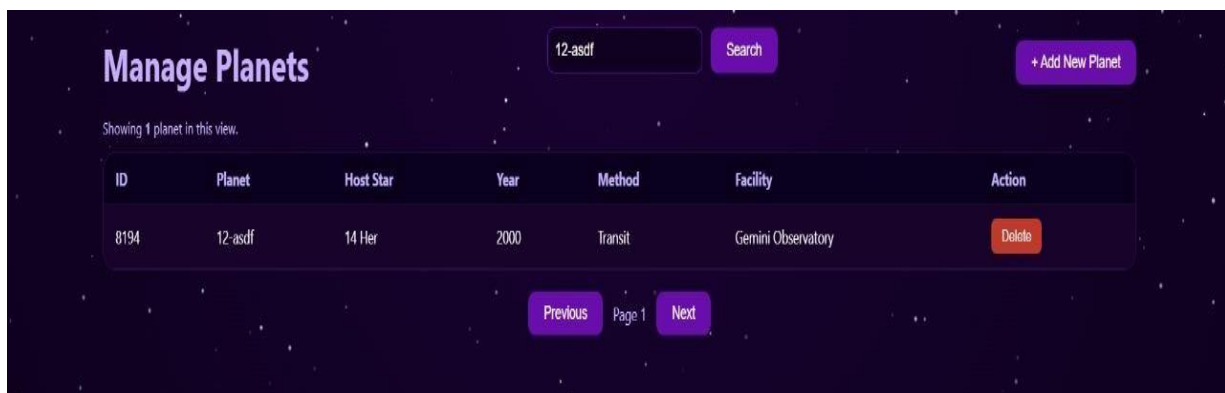
At the bottom right, there are two buttons: 'Cancel' and 'Save Planet'.

If a new planet is added, the page displays a message stating that the planet has been added successfully.



The 'Manage Planets' page features a search bar with the text '12-asdf' and a 'Search' button. A '+ Add New Planet' button is in the top right. A success message is displayed in a green box: 'Planet "12-asdf" added successfully (ID 8194)'. Below the message, it says 'Showing 100 planets in this view.' A table with 7 columns (ID, Planet, Host Star, Year, Method, Facility, Action) is partially visible.

The page allows admin to search for a specific planet by name, and displays the matching planet along with its details in the results.

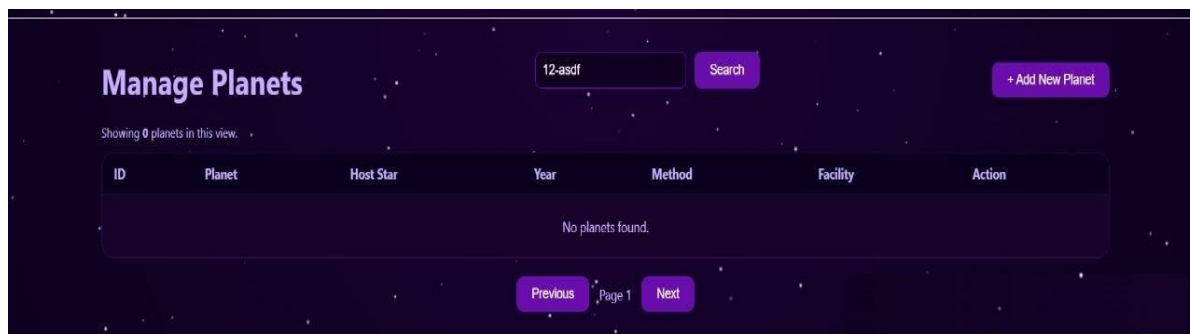
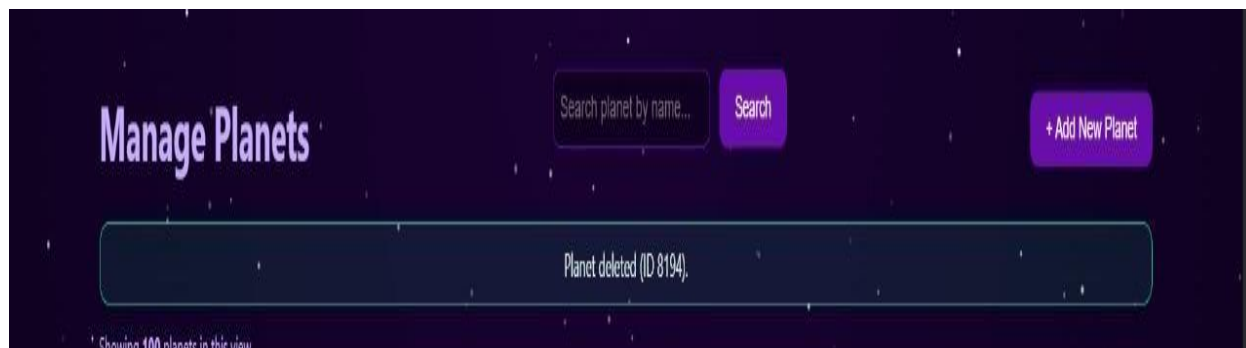


The 'Manage Planets' page shows the search results for '12-asdf'. The search bar contains '12-asdf' and the 'Search' button. The '+ Add New Planet' button is in the top right. A message states 'Showing 1 planet in this view.' The table displays one result:

ID	Planet	Host Star	Year	Method	Facility	Action
8194	12-asdf	14 Her	2000	Transit	Gemini Observatory	Delete

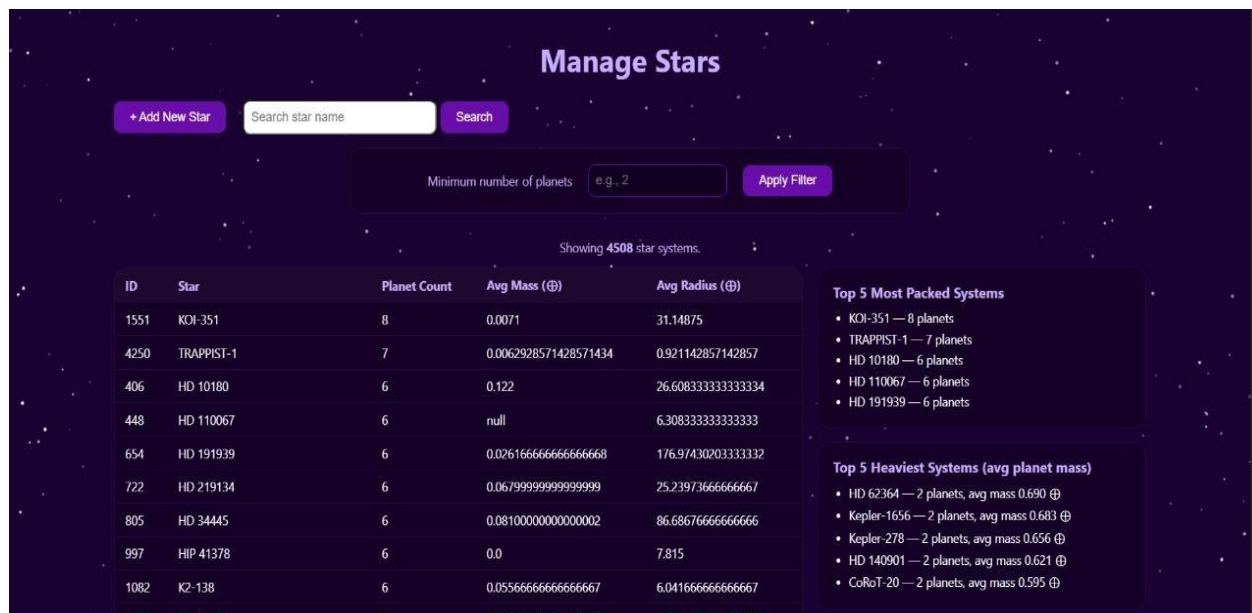
At the bottom, there are 'Previous', 'Page 1', and 'Next' buttons.

Admin can also delete the planet details.



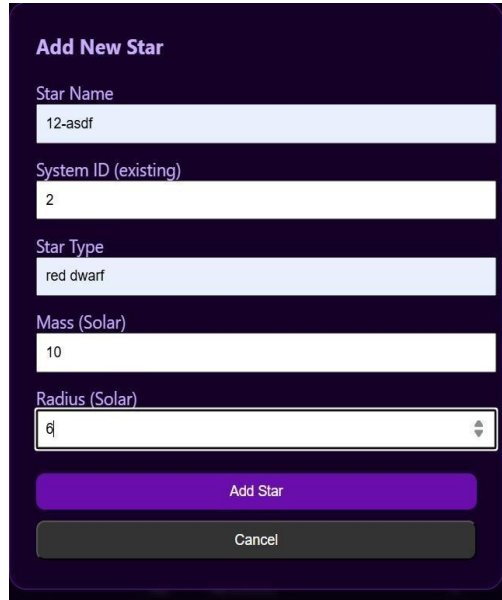
9. Manage Stars:

Description: The **Manage Stars** page allows admin to view, search, and filter star systems based on criteria like star name and minimum planet count. It displays a detailed table showing each star's planet count, average planet mass, and radius, while also providing quick insights through lists of the most packed and heaviest planetary systems. Users can additionally add new stars to the system through the provided interface.



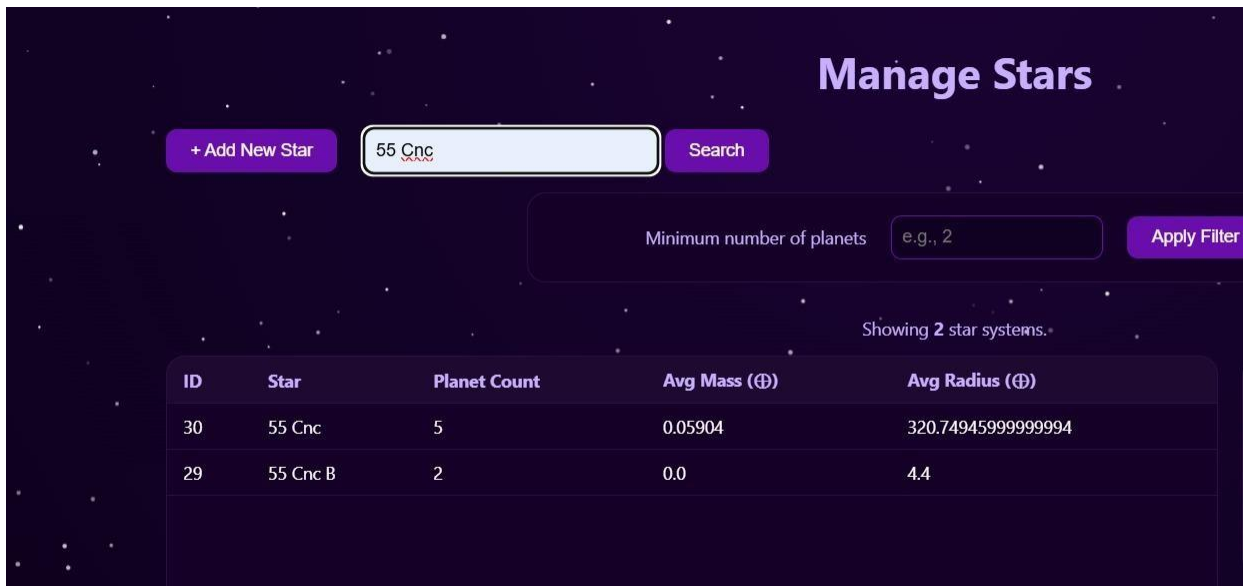
10. Add New Star:

Description: This page allows admin to add a new star to the system by filling out a form with the star's details. admin can enter the star's name, link it to an existing system through its System ID, specify the star type, and provide its mass and radius in solar units. After entering all required information, the user can click **Add Star** to save the new star to the database or select **Cancel** to discard the entry and return without making changes.



The 'Add New Star' form is a vertical stack of input fields on a dark purple background. It includes fields for 'Star Name' (containing '12-asdf'), 'System ID (existing)' (containing '2'), 'Star Type' (containing 'red dwarf'), 'Mass (Solar)' (containing '10'), and 'Radius (Solar)' (containing '6'). At the bottom are two buttons: a green 'Add Star' button and a grey 'Cancel' button.

When we search for a star by name, this page displays the number of planets orbiting the star, along with the star's average planet mass and average planet radius.



The 'Manage Stars' interface features a dark purple background with a starry pattern. At the top right is the title 'Manage Stars'. Below it is a search bar with a green '+ Add New Star' button on the left, a search input field containing '55 Cnc', and a green 'Search' button. Below the search bar is a filter section with the text 'Minimum number of planets' followed by an input field containing 'e.g., 2' and a green 'Apply Filter' button. Below the filter section, it says 'Showing 2 star systems.' and displays a table with the following data:

ID	Star	Planet Count	Avg Mass (⊕)	Avg Radius (⊕)
30	55 Cnc	5	0.05904	320.74945999999994
29	55 Cnc B	2	0.0	4.4

When a minimum number of planets is entered, the system filters and displays only those stars that have at least that number of planets.

Manage Stars

+ Add New Star Search star name Search

Minimum number of planets: 7 Apply Filter

Showing 2 star systems.

ID	Star	Planet Count	Avg Mass (⊕)	Avg Radius (⊕)
1551	KOI-351	8	0.0071	31.14875
4250	TRAPPIST-1	7	0.0062928571428571434	0.921142857142857

Top 5 Most Packed Systems

- KOI-351 — 8 planets
- TRAPPIST-1 — 7 planets
- HD 10180 — 6 planets
- HD 110067 — 6 planets
- HD 191939 — 6 planets

Top 5 Heaviest Systems (avg planet mass)

- HD 62364 — 2 planets, avg mass 0.690 ⊕
- Kepler-1656 — 2 planets, avg mass 0.683 ⊕
- Kepler-278 — 2 planets, avg mass 0.656 ⊕
- HD 140901 — 2 planets, avg mass 0.621 ⊕
- CoRoT-20 — 2 planets, avg mass 0.595 ⊕

11. View Logs:

Description: The **View Logs** page allows administrators to monitor and audit all recent system activities. It provides filters for username, action type, and date range, enabling precise retrieval of specific log records. The log table displays up to 300 entries, including the time of the action, the user who performed it, the action type, the entity involved (such as star, planet, user, or paper), the entity ID, and descriptive notes. This helps track events like logins, additions, updates, and deletions performed through the admin panel. The page also highlights the top five most active admins in the last seven days and shows paper upload activity from the past ten days, offering administrators a quick summary of recent engagement within the system.

View Logs

Home Search

Username: Action: -- Any -- From date: mm/dd/yyyy To date: mm/dd/yyyy

Apply Filters

Showing 75 log entries (max 300).

Time	User	Action	Entity	Entity ID	Notes
2025-12-03 12:59:55	Lira	LOGIN	user	1	User logged in
2025-12-03 12:58:43	Fiaa	LOGIN	user	4	User logged in
2025-12-03 12:55:24	Lira	DELETE_PLANET	planet	8196	Planet deleted from admin panel
2025-12-03 12:55:03	Lira	ADD_PLANET	planet	8196	Planet added via admin panel: 11 kLm
2025-12-03 12:52:52	Lira	LOGIN	user	1	User logged in
2025-12-03 05:12:48	Lira	LOGIN	user	1	User logged in
2025-12-03 05:07:37	Lira	DELETE_PAPER	paper	9	Deleted paper id=9
2025-12-03 05:07:34	Lira	DELETE_PAPER	paper	10	Deleted paper id=10
2025-12-03 05:07:14	Lira	ADD_PAPER	paper	13	Uploaded paper for planetid=13, title="kjh", fileName=FinalLab.pdf
2025-12-03 05:07:07	Lira	ADD_PAPER	paper	13	Uploaded paper for planetid=13, title="kjh", fileName=FinalLab.pdf

Top 5 admins (last 7 days)

- Lira — 55 actions
- Fiaa — 12 actions
- Tahsina — 3 actions
- Fia — 1 actions

Paper uploads per day (last 10 days)

- Dec 02 — 2 uploads

When a username is entered and the **LOGIN** action is selected, the system filters the logs and displays only the login activities performed by that specific user. After applying the filter, the table shows all matching entries, including the exact time of each login, the user’s name, the action type (LOGIN), the related entity (user), the entity ID, and notes confirming that the user successfully logged in. Only records that match both the username and the selected action appear, giving a clear and focused view of that user’s login history.

View Logs

Username

Fiaa

Action

LOGIN

From date

12/03/2025

To date

mm/dd/yyyy

Apply Filters

Showing 3 log entries (max 300).

Time	User	Action	Entity	Entity ID	Notes
2025-12-03 12:58:43	Fiaa	LOGIN	user	4	User logged in
2025-12-03 04:23:55	Fiaa	LOGIN	user	4	User logged in
2025-12-03 02:09:15	Fiaa	LOGIN	user	4	User logged in

Top 5 admins (last 7 days)

- Lira — 55 actions
- Fiaa — 12 actions
- Tahsina — 3 actions
- Fia — 1 actions

Paper uploads per day (last 10 days)

- Dec 02 — 2 uploads

When the username “**Lira**” is entered and the **ADD_PLANET** action is selected, the system filters the logs to display only the planet-adding activities performed by Lira. After applying the filter, the table shows all matching entries, including the exact timestamp of each action, the user’s name, the action type (ADD_PLANET), the affected entity (planet), the entity ID of the added planet, and detailed notes describing the planet added through the admin panel. Only log records that match both the selected user and the chosen action appear, giving a focused overview of all planets added by Lira.

Home Search

View Logs

Username

Lira

Action

ADD_PLANET

From date

mm/dd/yyyy

To date

mm/dd/yyyy

Apply Filters

Showing 5 log entries (max 300).

Time	User	Action	Entity	Entity ID	Notes
2025-12-03 12:55:03	Lira	ADD_PLANET	planet	8196	Planet added via admin panel: 11-klm
2025-12-03 02:19:22	Lira	ADD_PLANET	planet	8195	Planet added via admin panel: 24-poi
2025-12-02 20:12:49	Lira	ADD_PLANET	planet	8194	Planet added via admin panel: 12-asdf
2025-12-02 06:54:16	Lira	ADD_PLANET	planet	8193	Planet added via admin panel: 12-asdf
2025-12-02 02:45:18	Lira	ADD_PLANET	planet	8192	Planet added via admin panel: 24-asd

Top 5 admins (last 7 days)

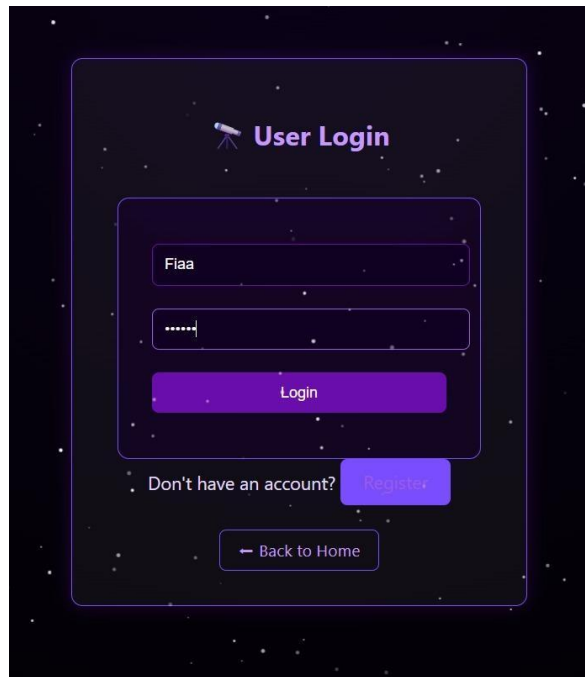
- Lira — 55 actions
- Fiaa — 12 actions
- Tahsina — 3 actions
- Fia — 1 actions

Paper uploads per day (last 10 days)

- Dec 02 — 2 uploads

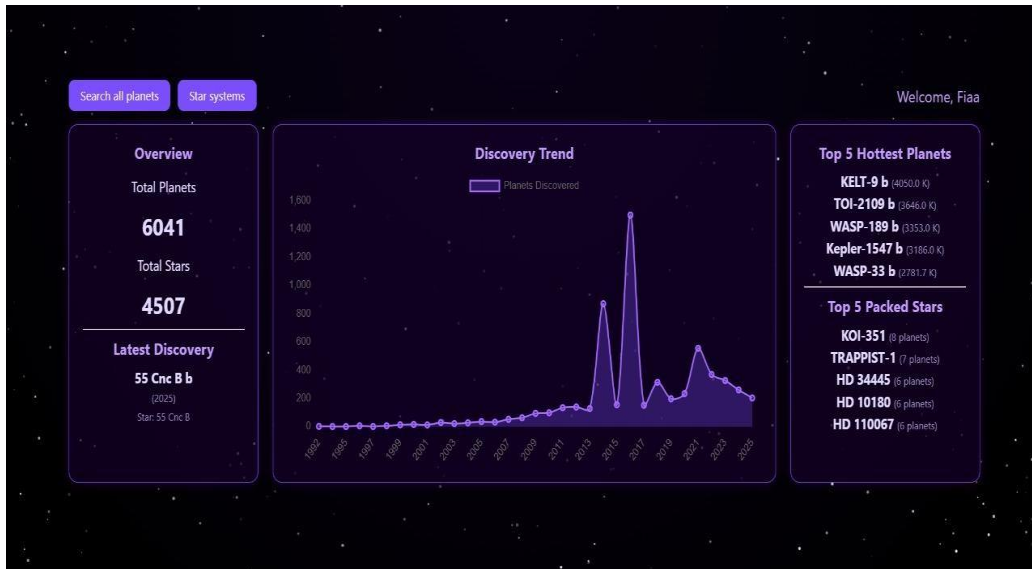
12. User login page:

Description: The User Login page allows registered users to securely access their accounts. It provides input fields for entering a valid **username** and **password**, which are authenticated upon clicking the **Login** button. If the credentials are correct, the user is redirected to their dashboard or designated homepage. The interface also offers a **Register** option for new users to create an account and a **Back to Home** button for returning to the main page. Overall, the page ensures a smooth and secure authentication process.



13. User Dashboard:

Description: This dashboard page provides a summarized and interactive view of planetary and stellar discoveries. At the top, users can navigate using buttons like “**Search all planets**” and “**Star systems**,” allowing quick access to detailed exploration features. The left section displays an **overview panel** showing the total number of discovered planets and stars, along with the **latest discovery** and its associated star. The center section presents a **Discovery Trend graph**, visualizing the number of planets discovered each year, helping users track scientific discovery patterns over time. On the right side, the page highlights the **Top 5 hottest planets** and **Top 5 most packed star systems**, offering quick insights into extreme or notable astronomical bodies. Additionally, a personalized welcome message appears at the top-right, enhancing user engagement. Overall, the page serves as a central hub for monitoring major statistics and trends related to planetary discovery.



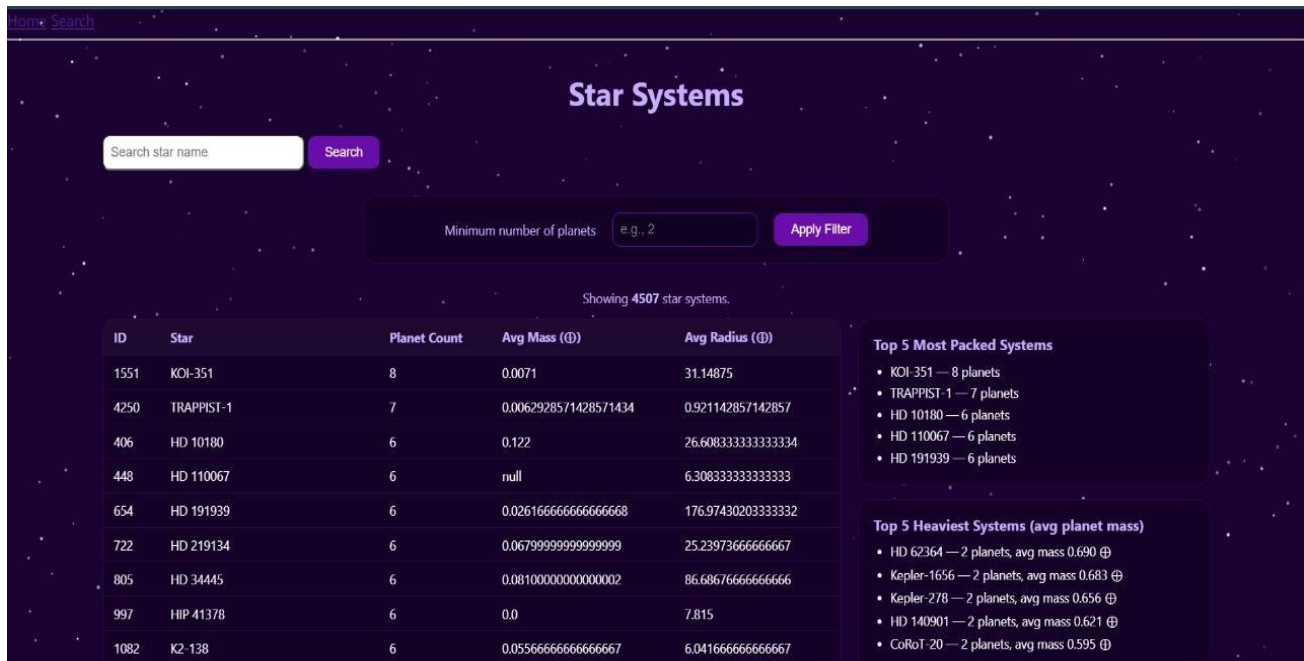
14. Search planet page

This page allows users to search, filter, and explore detailed information about discovered exoplanets. At the top, multiple search fields—**Planet or Star name**, **Year range (from–to)**, **Facility**, and **Discovery Method**—enable users to perform advanced filtering based on specific criteria. After entering the desired filters, users can click the **Search** button to display matching results. The main section presents a **structured table** listing each planet along with key scientific details such as its **host star**, **discovery year**, **method used**, **observatory/facility**, **distance in parsecs**, and **orbital period**. Each row includes a “**View**” button, allowing users to open a dedicated details page for deeper information about that specific planet. Overall, this page provides an organized and user-friendly interface for browsing planetary discovery data, supporting both quick lookups and detailed scientific exploration.

Search filters							
Planet or star	Year from	Year to	Method	Facility	Search		
Planet	Host Star	Year	Method	Facility	Distance (pc)	Period (days)	Details
55 Cnc B b	55 Cnc B	2025	Radial Velocity	Mauna Kea Observatory	12.4773	6.7994	View
55 Cnc B c	55 Cnc B	2025	Radial Velocity	Mauna Kea Observatory	12.4773	33.747	View
AU Mic e	AU Mic	2025	Radial Velocity	Mauna Kea Observatory	9.7221	33.11	View
Barnard c	Barnard's star	2025	Radial Velocity	Gemini Observatory	1.82655	4.1244	View
Barnard d	Barnard's star	2025	Radial Velocity	Gemini Observatory	1.82655	2.3402	View
Barnard e	Barnard's star	2025	Radial Velocity	Gemini Observatory	1.82655	6.7392	View
BD+05 4868 A b	BD+05 4868 A	2025	Transit	Transiting Exoplanet Survey Satellite (TESS)	43.5687	1.271869	View
BEBOp-3 b	BEBOp-3	2025	Radial Velocity	Haute-Provence Observatory	119.052	547.0	View
G 192-15 b	G 192-15	2025	Radial Velocity	Calar Alto Observatory	9.53114	2.27476	View
G 192-15 c	G 192-15	2025	Radial Velocity	Calar Alto Observatory	9.53114	1219.0	View
G 261-6 b	G 261-6	2025	Radial Velocity	Calar Alto Observatory	10.6267	5.4536	View
G 268-110 b	G 268-110	2025	Radial Velocity	Calar Alto Observatory		1.43263	View
Gaia-4 b	Gaia-4	2025	Astrometry	European Space Agency (ESA) Gaia Satellite	73.4829	571.3	View
Gaia-5 b	Gaia-5	2025	Astrometry	European Space Agency (ESA) Gaia Satellite	41.1813	358.62	View
GJ 2126 b	GJ 2126	2025	Radial Velocity	La Silla Observatory	38.129	272.7	View
GJ 251 c	GJ 251	2025	Radial Velocity	Multiple Observatories	5.58057	53.647	View
GJ 341 a	GJ 341	2025	Transit	Transiting Exoplanet Survey Satellite (TESS)	10.454	7.57686	View
GJ 3998 d	GJ 3998	2025	Radial Velocity	Multiple Observatories	18.1459	41.78	View

15. Star System Page:

This page allows users to explore and analyze different star systems along with the planets orbiting them. At the top, users can search for a specific star by typing its name in the **Search bar**, making it easy to quickly locate a known system. Additionally, a **filter option** lets users specify a minimum number of planets (e.g., 2 or more), and clicking **Apply Filter** displays only those systems that meet the selected criteria. The main table lists all star systems, showing essential information such as the **star ID**, **star name**, **number of orbiting planets**, **average planet mass**, and **average planet radius**. This helps users compare systems based on their planetary characteristics. The page also highlights two useful summaries on the right side: **Top 5 Most Packed Systems**, which ranks stars with the highest number of planets, and **Top 5 Heaviest Systems**, listing stars with planets having the highest average mass. Overall, the page provides a structured and interactive way to study the distribution, size, and mass characteristics of star systems, supporting both basic browsing and deeper scientific analysis.



The screenshot shows a web interface for "Star Systems". At the top, there is a navigation bar with "Home" and "Search" links. Below this is a search bar with the placeholder "Search star name" and a "Search" button. A filter section below the search bar allows users to set a "Minimum number of planets" (e.g., 2) and an "Apply Filter" button. The main content area displays a table of star systems, showing 4507 systems. The table has columns for ID, Star, Planet Count, Avg Mass (⊕), and Avg Radius (⊕). To the right of the table, there are two summary boxes: "Top 5 Most Packed Systems" and "Top 5 Heaviest Systems (avg planet mass)".

ID	Star	Planet Count	Avg Mass (⊕)	Avg Radius (⊕)
1551	KOI-351	8	0.0071	31.14875
4250	TRAPPIST-1	7	0.00629285/14285/1434	0.92114285/14285/7
406	HD 10180	6	0.122	26.608333333333334
448	HD 110067	6	null	6.308333333333333
654	HD 191939	6	0.026166666666666668	176.97430203333332
722	HD 219134	6	0.06799999999999999	25.239736666666667
805	HD 34445	6	0.08100000000000002	86.68676666666666
997	HIP 41378	6	0.0	7.815
1082	K2-138	6	0.05566666666666667	6.041666666666667

Showing 4507 star systems.

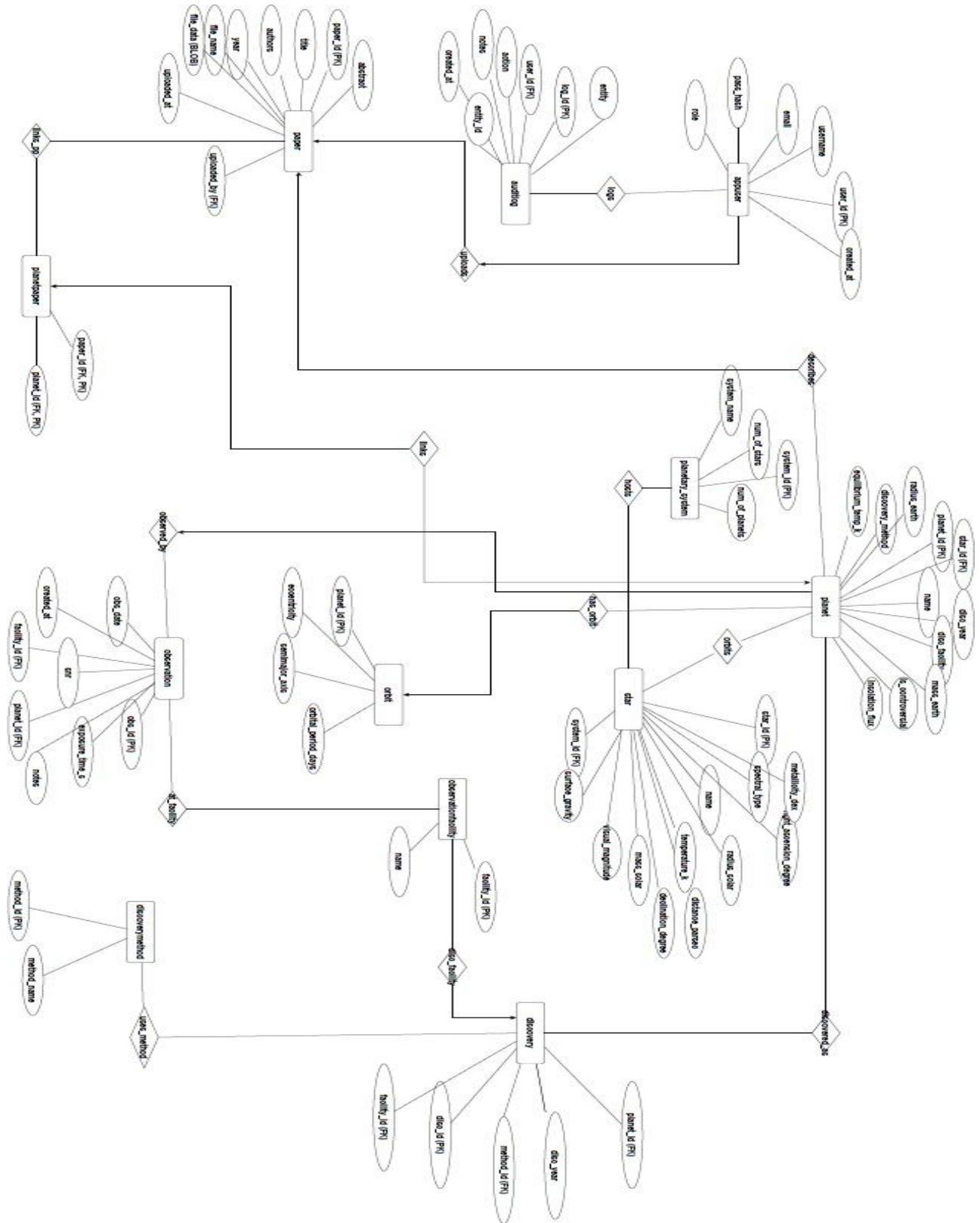
Top 5 Most Packed Systems

- KOI-351 — 8 planets
- TRAPPIST-1 — 7 planets
- HD 10180 — 6 planets
- HD 110067 — 6 planets
- HD 191939 — 6 planets

Top 5 Heaviest Systems (avg planet mass)

- HD 62364 — 2 planets, avg mass 0.690 ⊕
- Kepler-1656 — 2 planets, avg mass 0.683 ⊕
- Kepler-278 — 2 planets, avg mass 0.656 ⊕
- HD 140901 — 2 planets, avg mass 0.621 ⊕
- CoRoT-20 — 2 planets, avg mass 0.595 ⊕

Entity Relationship Diagram (Updated):



Narrative Description of ER-Diagram:

Entities:

AppUser

The *AppUser* entity represents all users registered in the system, including administrators and researchers. Each user is identified by a *User_ID* (primary key) and stores attributes such as *Username*, *Email*, *Password_Hash*, and *Role*. Users perform actions in the system, upload research papers, and generate audit logs.

AuditLog

The *AuditLog* entity captures every significant user action for monitoring and security. Identified by *Log_ID* (primary key), it stores the *Action_Type*, *Timestamp*, *Entity_Name*, *Entity_ID*, and *Notes*. Each log entry is linked to a specific user via *User_ID* (foreign key), enabling full traceability of user activity.

Paper

The *Paper* entity stores research papers related to exoplanets and star systems. Each paper is uniquely identified by *Paper_ID* (primary key) and includes attributes such as *Title*, *Authors*, *Year*, *Abstract*, and *FileName*. Papers are uploaded by users and may be linked to one or multiple planets through the *PlanetPaper* relationship.

Planetary System

The *Planetary_System* entity represents star systems containing one or more stars. Each system is identified by *System_ID* (primary key) and stores attributes like *System_Name* and *Number_of_Stars*. It serves as a parent entity for stars, providing organizational grouping for celestial structures.

PlanetPaper

The *PlanetPaper* entity is a junction table establishing a many-to-many relationship between planets and research papers. Identified by *PlanetPaper_ID* (primary key), it stores *Planet_ID* and *Paper_ID* (both foreign keys). This allows a single paper to reference multiple planets and a planet to be cited in multiple research papers.

Planet

The *Planet* entity contains detailed scientific information about exoplanets. Identified by *Planet_ID* (primary key), it stores *Name*, *Mass*, *Radius*, *Temperature*, *Discovery_Year*, and a reference to the *Star* it orbits. Each planet may have an associated orbit entry, discovery record, papers, and observations.

Star

The *Star* entity represents stars around which planets orbit. Each star is identified by *Star_ID* (primary key) and includes attributes like *Star_Name*, *Mass*, *Radius*, *Temperature*, and *Spectral_Type*. Every star belongs to a *Planetary_System* and may host multiple planets.

Orbit

The *Orbit* entity describes the orbital characteristics of each planet. Identified by *Orbit_ID* (primary key), it stores details like *Orbital_Period*, *SemiMajor_Axis*, and *Eccentricity*. It maintains a one-to-one relationship with a planet, defining the specific orbit parameters for that planet.

Observation

The *Observation* entity records scientific observations made for planets. Identified by *Observation_ID* (primary key), it stores *Observation_Date*, *Exposure_Time*, and *Data_Quality*. Each observation is linked to both a Planet and an *ObservationFacility*, documenting when and how the data was collected.

ObservationFacility

The *ObservationFacility* entity represents telescopes, observatories, or space instruments used for data collection. Identified by *Facility_ID* (primary key), it includes attributes such as *Facility_Name*, *Location*, and *Type* (e.g., *Satellite*, *Ground-based*). Facilities are connected to multiple observations and discoveries.

DiscoveryMethod

The *DiscoveryMethod* entity lists the formal methods used to detect exoplanets. Identified by *Method_ID* (primary key), it includes *Method_Name* and *Description* (e.g., *Transit*, *Radial Velocity*, *Direct Imaging*). Each discovery refers to one of these standardized detection techniques.

Discovery

The *Discovery* entity documents the official discovery information for each planet. Identified by *Discovery_ID* (primary key), it stores *Discovery_Date*, *Notes*, and references to the Planet, *DiscoveryMethod*, and *ObservationFacility* involved. It establishes the scientific context of how each planet was first identified.

Relationships:

Performs: An *AppUser* can perform multiple actions recorded in the AuditLog (1-to-Many).

Uploads: An *AppUser* can upload multiple Papers to the system (1-to-Many).

Cites: A *Paper* can reference multiple Planets, and a Planet can appear in multiple Papers through the PlanetPaper table (Many-to-Many).

Belongs To: A *Star* belongs to one Planetary_System, while a Planetary_System can contain multiple Stars (1-to-Many).

Hosts: A *Star* can host multiple Planets (1-to-Many).

Orbits: A Planet has exactly one Orbit record, describing its orbital parameters (1-to-1).

Observed At: A Planet can have multiple Observations, but each Observation is linked to only one Planet (1-to-Many).

Uses Facility: An ObservationFacility can support multiple Observations (1-to-Many).

Discovered Using: A DiscoveryMethod can be used for multiple Discoveries (1-to-Many).

Discovered By: A Planet has one Discovery record, while each Discovery is associated with exactly one Planet (1-to-1).

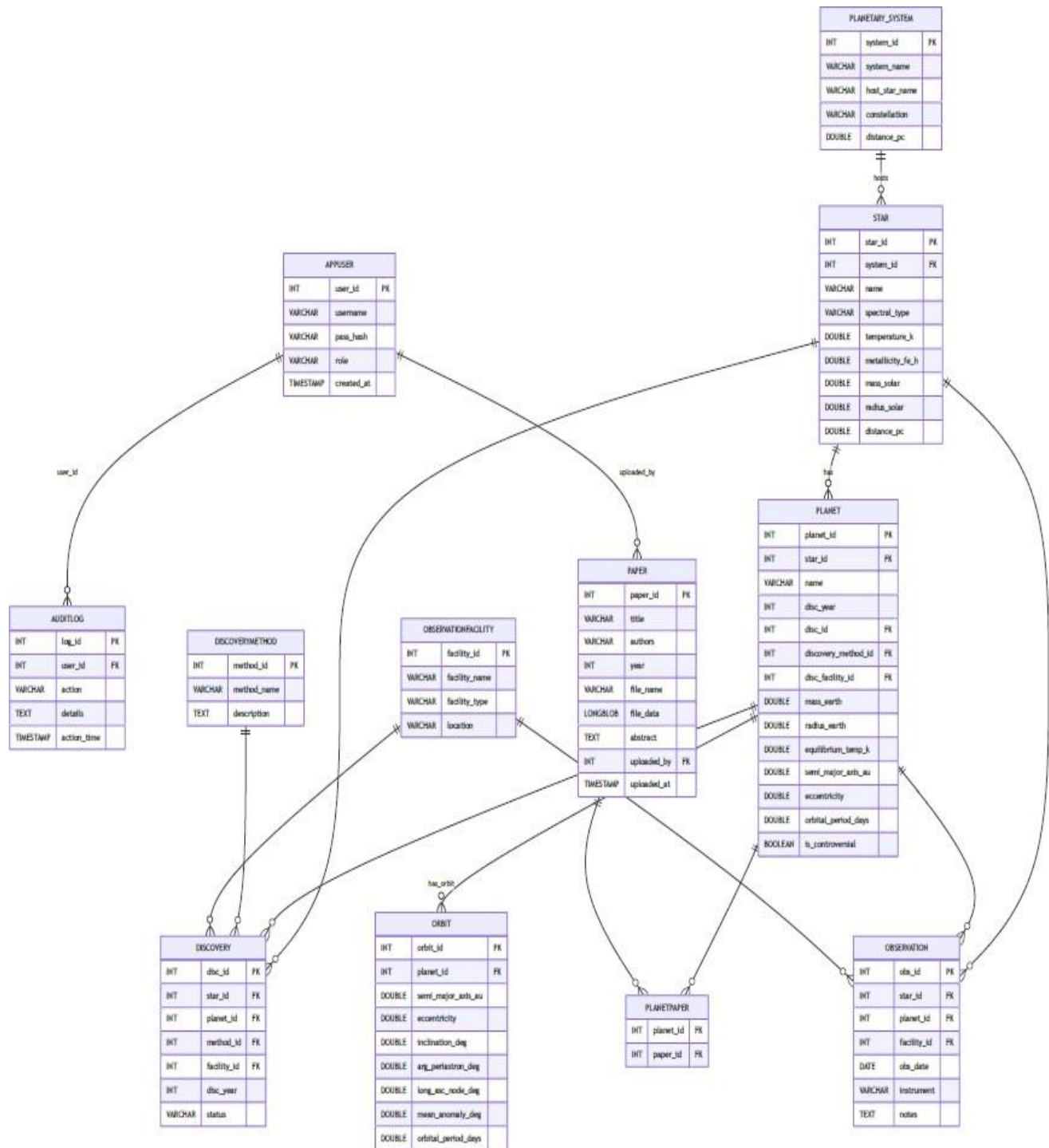
Facility Involved: An ObservationFacility can be associated with multiple Discoveries (1-to-Many).

System Has: A Planetary_System can include multiple Stars, and indirectly multiple Planets (1-to-Many).

Phase-02

Schema Refinement:

ER-Diagram converted into relational schema which is given below:



Integrity Constraints:

Table 1: AppUser

- Primary Key:
 - User_ID (Auto-increment)
- Unique Constraints:
 - Username must be unique
- Not Null:
 - Username, Password, Role
- Domain Constraints:
 - User_ID: Positive integer
 - Username: VARCHAR(100), non-empty
 - Password: VARCHAR(255), non-empty
 - Email: VARCHAR(255), valid email format
 - Role: ENUM('admin', 'researcher', 'viewer')

Table 2: AuditLog

- Primary Key:
 - Log_ID (Auto-increment)
- Foreign Keys:
 - User_ID REFERENCES AppUser(User_ID)
 - ON DELETE: SET NULL
 - ON UPDATE: CASCADE
- Not Null:
 - Action, Timestamp
- Domain Constraints:
 - Log_ID: Positive integer
 - Action: VARCHAR(100), non-empty
 - Timestamp: DATETIME
 - Entity: VARCHAR(50)
 - Entity_ID: Positive integer
 - Notes: TEXT

Table 3: Paper

- Primary Key:
 - Paper_ID (Auto-increment)
- Foreign Keys:
 - Uploaded_By REFERENCES AppUser(User_ID)
 - ON DELETE: SET NULL
 - ON UPDATE: CASCADE
- Not Null:
 - Title, Upload_Date

- Unique Constraints:
 - None specified
- Domain Constraints:
 - Paper_ID: Positive integer
 - Title: VARCHAR(255), non-empty
 - Year: INTEGER, ≥ 0
 - Upload_Date: DATE
 - File_Name: VARCHAR(255)

Table 4: Planetary System

- Primary Key:
 - System_ID (Auto-increment)
- Not Null:
 - Name
- Domain Constraints:
 - System_ID: Positive integer
 - Name: VARCHAR(255), non-empty
 - Star_Count: INTEGER, ≥ 0
 - Planet_Count: INTEGER, ≥ 0

Table 5: PlanetPaper (Associative Table)

- Primary Key:
 - PlanetPaper_ID (Auto-increment)
- Foreign Keys:
 - Planet_ID REFERENCES Planet(Planet_ID)
 - ON DELETE: CASCADE
 - ON UPDATE: CASCADE
 - Paper_ID REFERENCES Paper(Paper_ID)
 - ON DELETE: CASCADE
 - ON UPDATE: CASCADE
- Domain Constraints:
 - PlanetPaper_ID: Positive integer

Table 6: Planet

- Primary Key:
 - Planet_ID (Auto-increment)
- Foreign Keys:
 - Star_ID REFERENCES Star(Star_ID)
 - ON DELETE: SET NULL
 - ON UPDATE: CASCADE
 - Discovery_ID REFERENCES Discovery(Discovery_ID)
 - ON DELETE: SET NULL
 - ON UPDATE: CASCADE

- Not Null:
 - Name
- Domain Constraints:
 - Planet_ID: Positive integer
 - Name: VARCHAR(100), non-empty
 - Mass: DECIMAL(10,5), ≥ 0
 - Radius: DECIMAL(10,5), ≥ 0
 - Orbital_Period: DECIMAL(10,5), ≥ 0
 - Temperature: DECIMAL(10,2)

Table 7: Star

- Primary Key:
 - Star_ID (Auto-increment)
- Foreign Keys:
 - System_ID REFERENCES Planetary_System(System_ID)
 - ON DELETE: SET NULL
 - ON UPDATE: CASCADE
- Not Null:
 - Name
- Domain Constraints:
 - Star_ID: Positive integer
 - Name: VARCHAR(100), non-empty
 - Mass: DECIMAL(10,5), ≥ 0
 - Radius: DECIMAL(10,5), ≥ 0
 - Temperature: DECIMAL(10,2)

Table 8: Orbit

- Primary Key:
 - Orbit_ID (Auto-increment)
- Foreign Keys:
 - Planet_ID REFERENCES Planet(Planet_ID)
 - ON DELETE: CASCADE
 - ON UPDATE: CASCADE
- Not Null:
 - SemiMajor_Axis, Eccentricity
- Domain Constraints:
 - Orbit_ID: Positive integer
 - SemiMajor_Axis: DECIMAL(10,5), > 0
 - Eccentricity: DECIMAL(3,2) CHECK ($0 \leq \text{value} \leq 1$)

Table 9: Observation

- Primary Key:
 - Observation_ID (Auto-increment)

- Foreign Keys:
 - Planet_ID REFERENCES Planet(Planet_ID)
 - ON DELETE: CASCADE
 - ON UPDATE: CASCADE
 - Facility_ID REFERENCES ObservationFacility(Facility_ID)
 - ON DELETE: SET NULL
 - ON UPDATE: CASCADE
- Not Null:
 - Observation_Date
- Domain Constraints:
 - Observation_ID: Positive integer
 - Observation_Date: DATE
 - Details: TEXT

Table 10: ObservationFacility

- Primary Key:
 - Facility_ID (Auto-increment)
- Not Null:
 - Name, Location
- Domain Constraints:
 - Facility_ID: Positive integer
 - Name: VARCHAR(255), non-empty
 - Location: VARCHAR(255), non-empty

Table 11: DiscoveryMethod

- Primary Key:
 - Method_ID (Auto-increment)
- Not Null:
 - Method_Name
- Domain Constraints:
 - Method_ID: Positive integer
 - Method_Name: VARCHAR(100), non-empty

Table 12: Discovery

- Primary Key:
 - Discovery_ID (Auto-increment)
- Foreign Keys:
 - Method_ID REFERENCES DiscoveryMethod(Method_ID)
 - ON DELETE: SET NULL
 - ON UPDATE: CASCADE
 - Facility_ID REFERENCES ObservationFacility(Facility_ID)
 - ON DELETE: SET NULL
 - ON UPDATE: CASCADE

- Not Null:
 - Year
- Domain Constraints:
 - Discovery_ID: Positive integer
 - Year: INTEGER, ≥ 0
 - Notes: TEXT

Normalization :

1st Normal Form (1NF)

The given relation is in 1st Normal Form because it meets the following criteria:

1. All cells contain atomic (indivisible) values.
2. Entries in each column are of the same data type.
3. Each record is unique; there are no duplicate rows.
4. Attribute names are unique and meaningful.

2nd Normal Form (2NF)

The given relation is in 2nd Normal Form because it satisfies these conditions:

1. The table is already in 1st Normal Form.
2. There is no partial dependency; no subset of a candidate key determines a non-key attribute.
3. All non-key attributes are fully dependent on the entire primary key.

3rd Normal Form (3NF)

The given relation is in 3rd Normal Form because:

1. The table is already in 2nd Normal Form.
2. There is no transitive dependency; a non-prime attribute does not depend on another non-prime attribute.
3. For every functional dependency, the left-hand side is a key, and the right-hand side is a non-prime attribute.

Thus, all columns are determined only by the primary key and no other non-key attributes.

Normalization (1NF, 2NF, 3NF) – ExoDiscover Database

Appuser

```
mysql> DESCRIBE appuser;
```

Field	Type	Null	Key	Default	Extra
user_id	int	NO	PRI	NULL	auto_increment
username	varchar(50)	NO	UNI	NULL	
email	varchar(120)	NO	UNI	NULL	
pass_hash	varchar(100)	NO		NULL	
role	enum('admin','user')	NO		user	
created_at	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

6 rows in set (0.01 sec)

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	Table 'appuser' has atomic columns, no repeating groups, and each row is uniquely identified by its primary key.
Second Normal Form (2NF)	✓	Table 'appuser' is in 1NF and the primary key is a single attribute, so no partial dependency exists.
Third Normal Form (3NF)	✓	Table 'appuser' has no transitive dependencies—each non-key attribute depends only on the primary key.

Auditlog

```
mysql> DESCRIBE auditlog;
```

Field	Type	Null	Key	Default	Extra
log_id	bigint	NO	PRI	NULL	auto_increment
user_id	int	YES		NULL	
action	varchar(80)	NO		NULL	
entity	varchar(80)	YES		NULL	
entity_id	varchar(80)	YES		NULL	
notes	varchar(500)	YES		NULL	
created_at	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	Table 'auditlog' has atomic columns, no repeating groups, and each row is uniquely identified by its primary key.
Second Normal Form (2NF)	✓	Table 'auditlog' is in 1NF and the primary key is a single attribute, so no partial dependency exists.
Third Normal Form (3NF)	✓	Table 'auditlog' has no transitive dependencies—each non-key attribute depends only on the primary key.

Discovery

```
mysql> DESCRIBE discovery;
+-----+-----+-----+-----+-----+-----+
| Field      | Type | Null | Key | Default | Extra           |
+-----+-----+-----+-----+-----+-----+
| disc_id    | int  | NO   | PRI | NULL    | auto_increment |
| planet_id  | int  | NO   | MUL | NULL    |                 |
| method_id  | int  | YES  | MUL | NULL    |                 |
| facility_id| int  | YES  | MUL | NULL    |                 |
| disc_year  | int  | YES  | MUL | NULL    |                 |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	Table 'discovery' has atomic columns, no repeating groups, and each row is uniquely identified by its primary key.
Second Normal Form (2NF)	✓	Table 'discovery' is in 1NF and the primary key is a single attribute, so no partial dependency exists.
Third Normal Form (3NF)	✓	Table 'discovery' has no transitive dependencies—each non-key attribute depends only on the primary key.

Discoverymethod

```
mysql> DESCRIBE discoverymethod;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra           |
+-----+-----+-----+-----+-----+-----+
| method_id  | int           | NO   | PRI | NULL    | auto_increment |
| method_name | varchar(80)   | YES  | UNI | NULL    |                 |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	Table 'discoverymethod' has atomic columns, no repeating groups, and each row is uniquely identified by its primary key.
Second Normal Form (2NF)	✓	Table 'discoverymethod' is in 1NF and the primary key is a single attribute, so no partial dependency exists.
Third Normal Form (3NF)	✓	Table 'discoverymethod' has no transitive dependencies—each non-key attribute depends only on the primary key.

Observation

```
mysql> DESCRIBE observation;
```

Field	Type	Null	Key	Default	Extra
obs_id	int	NO	PRI	NULL	auto_increment
planet_id	int	NO	MUL	NULL	
obs_date	date	YES		NULL	
facility_id	int	YES	MUL	NULL	
notes	varchar(500)	YES		NULL	
exposure_time_s	double	YES		NULL	
snr	double	YES		NULL	
created_at	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	Table 'observation' has atomic columns, no repeating groups, and each row is uniquely identified by its primary key.
Second Normal Form (2NF)	✓	Table 'observation' is in 1NF and the primary key is a single attribute, so no partial dependency exists.
Third Normal Form (3NF)	✓	Table 'observation' has no transitive dependencies—each non-key attribute depends only on the primary key.

Observationfacility

```
mysql> DESCRIBE observationfacility;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra           |
+-----+-----+-----+-----+-----+-----+
| facility_id | int           | NO   | PRI | NULL    | auto_increment |
| name        | varchar(150)  | YES  | UNI | NULL    |                 |
+-----+-----+-----+-----+-----+-----+
```

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	Table 'observationfacility' has atomic columns, no repeating groups, and each row is uniquely identified by its primary key.
Second Normal Form (2NF)	✓	Table 'observationfacility' is in 1NF and the primary key is a single attribute, so no partial dependency exists.
Third Normal Form (3NF)	✓	Table 'observationfacility' has no transitive dependencies—each non-key attribute depends only on the primary key.

Orbit

```
mysql> DESCRIBE orbit;
+-----+-----+-----+-----+-----+-----+
| Field                | Type   | Null  | Key  | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| planet_id            | int    | NO    | PRI  | NULL    |       |
| orbital_period_days  | double | YES   |      | NULL    |       |
| semimajor_axis       | double | YES   |      | NULL    |       |
| eccentricity         | double | YES   |      | NULL    |       |
+-----+-----+-----+-----+-----+-----+
```

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	Table 'orbit' has atomic columns, no repeating groups, and each row is uniquely identified by its primary key.
Second Normal Form (2NF)	✓	Table 'orbit' is in 1NF and the primary key is a single attribute, so no partial dependency exists.
Third Normal Form (3NF)	✓	Table 'orbit' has no transitive dependencies—each non-key attribute depends only on the primary key.

Paper

```
mysql> DESCRIBE paper;
```

Field	Type	Null	Key	Default	Extra
paper_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
authors	varchar(300)	YES		NULL	
year	int	YES		NULL	
abstract	text	YES		NULL	
uploaded_by	int	YES	MUL	NULL	
uploaded_at	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED
file_name	varchar(300)	YES		NULL	
file_data	longblob	YES		NULL	

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	Table 'paper' has atomic columns, no repeating groups, and each row is uniquely identified by its primary key.
Second Normal Form (2NF)	✓	Table 'paper' is in 1NF and the primary key is a single attribute, so no partial dependency exists.
Third Normal Form (3NF)	✓	Table 'paper' has no transitive dependencies—each non-key attribute depends only on the primary key.

Planet

```
mysql> DESCRIBE planet;
```

Field	Type	Null	Key	Default	Extra
planet_id	int	NO	PRI	NULL	auto_increment
star_id	int	NO	MUL	NULL	
name	varchar(120)	YES	UNI	NULL	
mass_earth	double	YES		NULL	
radius_earth	double	YES		NULL	
equilibrium_temp_k	double	YES		NULL	
disc_year	int	YES		NULL	
discovery_method	varchar(100)	YES		NULL	
disc_facility	varchar(150)	YES		NULL	
is_controversial	tinyint(1)	YES		NULL	
insolation_flux	double	YES		NULL	
image_path	varchar(200)	YES		NULL	

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	Table 'planet' has atomic columns, no repeating groups, and each row is uniquely identified by its primary key.
Second Normal Form (2NF)	✓	Table 'planet' is in 1NF and the primary key is a single attribute, so no partial dependency exists.
Third Normal Form (3NF)	✓	Table 'planet' has no transitive dependencies—each non-key attribute depends only on the primary key.

planetary_system

```
mysql> DESCRIBE planetary_system;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| system_id      | int           | NO   | PRI | NULL    | auto_increment |
| system_name    | varchar(120)  | YES  | UNI | NULL    |                |
| num_of_stars   | int           | YES  |     | NULL    |                |
| num_of_planets | int           | YES  |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	Table 'planetary_system' has atomic columns, no repeating groups, and each row is uniquely identified by its primary key.
Second Normal Form (2NF)	✓	Table 'planetary_system' is in 1NF and the primary key is a single attribute, so no partial dependency exists.
Third Normal Form (3NF)	✓	Table 'planetary_system' has no transitive dependencies—each non-key attribute depends only on the primary key.

Planetpaper

```
mysql> DESCRIBE planetpaper;
```

```
+-----+-----+-----+-----+-----+-----+
| Field      | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| planet_id  | int  | NO   | PRI | NULL    |       |
| paper_id   | int  | NO   | PRI | NULL    |       |
+-----+-----+-----+-----+-----+-----+
```

```
2 rows in set (0.00 sec)
```

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	Table 'planetpaper' has atomic columns, no repeating groups, and each row is uniquely identified by its primary key.
Second Normal Form (2NF)	✓	Table 'planetpaper' is in 1NF and the primary key is a single attribute, so no partial dependency exists.
Third Normal Form (3NF)	✓	Table 'planetpaper' has no transitive dependencies—each non-key attribute depends only on the primary key.

Star

```
mysql> DESCRIBE star;
+-----+-----+-----+-----+-----+-----+
| Field                | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| star_id              | int           | NO   | PRI | NULL    | auto_increment |
| system_id            | int           | NO   | MUL | NULL    |                |
| name                 | varchar(120)  | YES  | UNI | NULL    |                |
| spectral_type        | varchar(20)   | YES  |     | NULL    |                |
| temperature_k        | double        | YES  |     | NULL    |                |
| radius_solar         | double        | YES  |     | NULL    |                |
| mass_solar           | double        | YES  |     | NULL    |                |
| metallicity_dex      | double        | YES  |     | NULL    |                |
| surface_gravity      | double        | YES  |     | NULL    |                |
| right_ascension_degree | double        | YES  |     | NULL    |                |
| declination_degree   | double        | YES  |     | NULL    |                |
| distance_parsec      | double        | YES  |     | NULL    |                |
| visual_magnitude     | double        | YES  |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
13 rows in set (0.00 sec)
```

Normal Form	✓ / ✗	Explanation
First Normal Form (1NF)	✓	Table 'star' has atomic columns, no repeating groups, and each row is uniquely identified by its primary key.
Second Normal Form (2NF)	✓	Table 'star' is in 1NF and the primary key is a single attribute, so no partial dependency exists.
Third Normal Form (3NF)	✓	Table 'star' has no transitive dependencies—each non-key attribute depends only on the primary key.

SQL Statements:

The necessary SQL commands required to create the database 'exoplanet_db;' with appropriate foreign keys are shown below:

Creation of exoplanet_db database:

Create database exoplanet_db;

Use exoplanet_db;

Creation of PLANETARY SYSTEM table:

```
CREATE TABLE planetary_system (  
    system_id    INT NOT NULL AUTO_INCREMENT,  
    system_name  VARCHAR(120) UNIQUE,  
    num_of_stars INT,  
    num_of_planets INT,  
    PRIMARY KEY (system_id)  
);
```

2) Creation of STAR table:

```
CREATE TABLE star (  
    star_id      INT NOT NULL AUTO_INCREMENT,  
    system_id    INT NOT NULL,  
    name         VARCHAR(120) UNIQUE,  
    spectral_type VARCHAR(20),  
    temperature_k DOUBLE,  
    radius_solar  DOUBLE,  
    mass_solar    DOUBLE,  
    metallicity_dex DOUBLE,  
    surface_gravity DOUBLE,  
    right_ascension_degree DOUBLE,  
    declination_degree DOUBLE,
```

```

distance_parsec    DOUBLE,
visual_magnitude   DOUBLE,
PRIMARY KEY (star_id),
KEY fk_star_system (system_id),
CONSTRAINT fk_star_system
    FOREIGN KEY (system_id) REFERENCES planetary_system(system_id)
);

```

3) Creation Of PLANET Table

```

CREATE TABLE planet (
    planet_id      INT NOT NULL AUTO_INCREMENT,
    star_id        INT NOT NULL,
    name           VARCHAR(120) NOT NULL UNIQUE,
    mass_earth     DOUBLE,
    radius_earth   DOUBLE,
    equilibrium_temp_k DOUBLE,
    disc_year      INT,
    discovery_method VARCHAR(100),
    disc_facility   VARCHAR(150),
    is_controversial TINYINT(1),
    insolation_flux DOUBLE,
    image_path      VARCHAR(200),
    PRIMARY KEY (planet_id),
    KEY fk_planet_star (star_id),
    CONSTRAINT fk_planet_star
        FOREIGN KEY (star_id) REFERENCES star(star_id)
);

```

4) Creation Of Orbit Table

```
CREATE TABLE orbit (  
    planet_id    INT    NOT NULL,  
    orbital_period_days DOUBLE,  
    semimajor_axis    DOUBLE,  
    eccentricity    DOUBLE,  
    PRIMARY KEY (planet_id),  
    CONSTRAINT fk_orbit_planet  
        FOREIGN KEY (planet_id) REFERENCES planet(planet_id)  
);
```

5) Creation Of DISCOVERY METHOD Table

```
CREATE TABLE discoverymethod (  
    method_id INT NOT NULL AUTO_INCREMENT,  
    method_name VARCHAR(80) UNIQUE,  
    PRIMARY KEY (method_id)  
);
```

6) Creation Of OBSERVATION FACILITY Table

```
CREATE TABLE observationfacility (  
    facility_id INT NOT NULL AUTO_INCREMENT,  
    name    VARCHAR(150) UNIQUE,  
    PRIMARY KEY (facility_id)  
);
```

7) Creation Of DISCOVERY Table

```
CREATE TABLE discovery (  
    disc_id INT NOT NULL AUTO_INCREMENT,  
    planet_id INT NOT NULL,  
    method_id INT NOT NULL,  
    facility_id INT,  
    disc_year INT,  
    PRIMARY KEY (disc_id),  
    KEY fk_disc_planet (planet_id),  
    KEY fk_disc_method (method_id),  
    KEY fk_disc_facility (facility_id),  
    CONSTRAINT fk_disc_planet  
        FOREIGN KEY (planet_id) REFERENCES planet(planet_id),  
    CONSTRAINT fk_disc_method  
        FOREIGN KEY (method_id) REFERENCES discoverymethod(method_id),  
    CONSTRAINT fk_disc_facility  
        FOREIGN KEY (facility_id) REFERENCES observationfacility(facility_id)  
);
```

8) Creation Of OBSERVATION Table

```
CREATE TABLE observation (  
    obs_id INT NOT NULL AUTO_INCREMENT,  
    planet_id INT NOT NULL,  
    obs_date DATE,  
    facility_id INT,  
    notes VARCHAR(500),  
    exposure_time_s DOUBLE,
```

```

snr    DOUBLE,
created_at    TIMESTAMP NULL DEFAULT CURRENT_TIMESTAMP,
PRIMARY KEY (obs_id),
KEY fk_obs_planet (planet_id),
KEY fk_obs_facility (facility_id),
CONSTRAINT fk_obs_planet
    FOREIGN KEY (planet_id) REFERENCES planet(planet_id),
CONSTRAINT fk_obs_facility
    FOREIGN KEY (facility_id) REFERENCES observationfacility(facility_id)
);

```

9) Creation Of APPUSER Table(users & admins)

```

CREATE TABLE appuser (
    user_id INT NOT NULL AUTO_INCREMENT,
    username VARCHAR(50) NOT NULL UNIQUE,
    email VARCHAR(120) NOT NULL UNIQUE,
    pass_hash VARCHAR(100) NOT NULL,
    role ENUM('admin','user') NOT NULL DEFAULT 'user',
    created_at TIMESTAMP NULL DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (user_id)
);

```

10) Creation Of PAPER Table(uploaded research papers)

```

CREATE TABLE paper (
    paper_id INT NOT NULL AUTO_INCREMENT,
    title VARCHAR(200) NOT NULL,
    authors VARCHAR(300),
    year INT,

```

```

abstract    TEXT,
uploaded_by INT,
uploaded_at TIMESTAMP NULL DEFAULT CURRENT_TIMESTAMP,
file_name   VARCHAR(300),
file_data   LONGBLOB,
PRIMARY KEY (paper_id),
KEY fk_paper_user (uploaded_by),
CONSTRAINT fk_paper_user
    FOREIGN KEY (uploaded_by) REFERENCES appuser(user_id)
);

```

11) Creation Of PLANET–PAPER JUNCTION TABLE

```

CREATE TABLE planetpaper (
    planet_id INT NOT NULL,
    paper_id INT NOT NULL,
    PRIMARY KEY (planet_id,paper_id),
    KEY fk_pp_paper (paper_id),
    CONSTRAINT fk_pp_planet
        FOREIGN KEY (planet_id) REFERENCES planet(planet_id),
    CONSTRAINT fk_pp_paper
        FOREIGN KEY (paper_id) REFERENCES paper(paper_id)
);

```

12) Creation Of AUDIT LOG Table

```

CREATE TABLE auditlog (
    log_id    BIGINT NOT NULL AUTO_INCREMENT,
    user_id   INT,
    action    VARCHAR(80) NOT NULL,
    entity    VARCHAR(80),
    entity_id VARCHAR(80),

```

```
notes    VARCHAR(500),
created_at TIMESTAMP NULL DEFAULT CURRENT_TIMESTAMP,
PRIMARY KEY (log_id),
KEY fk_log_user (user_id),
CONSTRAINT fk_log_user
    FOREIGN KEY (user_id) REFERENCES appuser(user_id)
);
```

Populating the Database:

Set of data have been inserted into each relation of the created database `exoplanet_db`.

After insertion of data, the tables are currently in this form:

Discovery:

```
mysql> SELECT * FROM discovery ORDER BY disc_id ASC limit 10;
```

disc_id	planet_id	method_id	facility_id	disc_year
1	1	1	1	2007
2	2	1	2	2009
3	3	1	3	2008
4	4	1	4	2002
5	5	1	5	1996
6	6	1	6	2020
7	7	1	3	2008
8	8	2	7	2008
9	9	1	3	2018
10	10	1	6	2010

Auditlog:

```
mysql> select * from auditlog order by log_id limit 30;
```

log_id	user_id	action	entity	entity_id	notes	created_at
1	1	LOGIN	user	1	User logged in	2025-11-23 23:48:10
2	1	LOGIN	user	1	User logged in	2025-11-23 23:51:20
3	1	LOGIN	user	1	User logged in	2025-11-24 00:01:59
4	1	LOGIN	user	1	User logged in	2025-11-24 00:05:19
5	1	LOGIN	user	1	User logged in	2025-11-26 20:15:59
6	1	LOGIN	user	1	User logged in	2025-11-27 11:59:30
7	2	LOGIN	user	2	User logged in	2025-11-27 13:35:37
8	1	LOGIN	user	1	User logged in	2025-11-27 13:36:01
9	1	LOGIN	user	1	User logged in	2025-11-29 02:15:24
10	1	LOGIN	user	1	User logged in	2025-11-29 02:37:12
11	1	LOGIN	user	1	User logged in	2025-11-29 02:42:25
12	1	LOGIN	user	1	User logged in	2025-11-29 02:43:37
13	1	LOGIN	user	1	User logged in	2025-11-29 02:51:39
14	1	LOGIN	user	1	User logged in	2025-11-29 03:04:14
15	1	LOGIN	user	1	User logged in	2025-11-29 03:29:30
16	1	LOGIN	user	1	User logged in	2025-11-30 15:34:58
17	1	LOGIN	user	1	User logged in	2025-12-01 12:40:52
18	1	LOGIN	user	1	User logged in	2025-12-01 18:47:37
19	3	LOGIN	user	3	User logged in	2025-12-01 18:48:36
20	1	LOGIN	user	1	User logged in	2025-12-01 18:49:20
21	1	LOGIN	user	1	User logged in	2025-12-01 19:31:04
22	1	LOGIN	user	1	User logged in	2025-12-01 20:12:42
23	1	DELETE_PLANET	planet	54	Planet deleted from admin panel	2025-12-01 20:12:57
24	1	LOGIN	user	1	User logged in	2025-12-01 20:39:26
25	1	LOGIN	user	1	User logged in	2025-12-01 20:42:14
26	1	ADD_PLANET	planet	8192	Planet added via admin panel: 24-asd	2025-12-01 20:45:18
27	1	DELETE_PLANET	planet	8192	Planet deleted from admin panel	2025-12-01 20:48:19
28	1	LOGIN	user	1	User logged in	2025-12-02 00:53:34
29	1	ADD_PLANET	planet	8193	Planet added via admin panel: 12-asdf	2025-12-02 00:54:16
30	1	DELETE_PLANET	planet	8193	Planet deleted from admin panel	2025-12-02 00:54:54

Discoverymethod:

```
mysql> select * from discoverymethod order by method_id asc;
```

method_id	method_name
1	Radial Velocity
2	Imaging
3	Eclipse Timing Variations
4	Microlensing
5	Transit
6	Transit Timing Variations
7	Astrometry
8	Disk Kinematics
9	Orbital Brightness Modulation
10	Pulsation Timing Variations
11	Pulsar Timing

Planet:

```
mysql> select * from orbit order by planet_id limit 20;
```

planet_id	orbital_period_days	semimajor_axis	eccentricity
1	323.21	1.178	12.2
2	516.21997	1.53	12.3
3	186.76	0.775	13.1
4	1765.0389	2.774	12.6
5	798.5	1.66	13.5
6	578.38	1.45	12.9
7	982.85	2.476	12.5
9	30.33	0.194	13.9
10	452.8	1.333	13.4
11	883	2.08	13.9
22	406	0.92	13.4
25	335.10001	0.99	12.3
26	270.27	0.877	12.6
27	1078	2.1	13.2
28	2391	3.6	14.2
29	14002	11.6	13.5
30	9100	10.4	NULL
31	4.230785	0.0527	14.3
32	6.7994	0.044	1.69
33	33.747	0.13	2.16

```
20 rows in set (0.00 sec)
```

Auditlog:

```
mysql> select * from auditlog order by log_id limit 30;
```

log_id	user_id	action	entity	entity_id	notes	created_at
1	1	LOGIN	user	1	User logged in	2025-11-23 23:48:10
2	1	LOGIN	user	1	User logged in	2025-11-23 23:51:20
3	1	LOGIN	user	1	User logged in	2025-11-24 00:01:59
4	1	LOGIN	user	1	User logged in	2025-11-24 00:05:19
5	1	LOGIN	user	1	User logged in	2025-11-26 20:15:59
6	1	LOGIN	user	1	User logged in	2025-11-27 11:59:30
7	2	LOGIN	user	2	User logged in	2025-11-27 13:35:37
8	1	LOGIN	user	1	User logged in	2025-11-27 13:36:01
9	1	LOGIN	user	1	User logged in	2025-11-29 02:15:24
10	1	LOGIN	user	1	User logged in	2025-11-29 02:37:12
11	1	LOGIN	user	1	User logged in	2025-11-29 02:42:25
12	1	LOGIN	user	1	User logged in	2025-11-29 02:43:37
13	1	LOGIN	user	1	User logged in	2025-11-29 02:51:39
14	1	LOGIN	user	1	User logged in	2025-11-29 03:04:14
15	1	LOGIN	user	1	User logged in	2025-11-29 03:29:30
16	1	LOGIN	user	1	User logged in	2025-11-30 15:34:58
17	1	LOGIN	user	1	User logged in	2025-12-01 12:40:52
18	1	LOGIN	user	1	User logged in	2025-12-01 18:47:37
19	3	LOGIN	user	3	User logged in	2025-12-01 18:48:36
20	1	LOGIN	user	1	User logged in	2025-12-01 18:49:20
21	1	LOGIN	user	1	User logged in	2025-12-01 19:31:04
22	1	LOGIN	user	1	User logged in	2025-12-01 20:12:42
23	1	DELETE_PLANET	planet	54	Planet deleted from admin panel	2025-12-01 20:12:57
24	1	LOGIN	user	1	User logged in	2025-12-01 20:39:26
25	1	LOGIN	user	1	User logged in	2025-12-01 20:42:14
26	1	ADD_PLANET	planet	8192	Planet added via admin panel: 24-asd	2025-12-01 20:45:18
27	1	DELETE_PLANET	planet	8192	Planet deleted from admin panel	2025-12-01 20:48:19
28	1	LOGIN	user	1	User logged in	2025-12-02 00:53:34
29	1	ADD_PLANET	planet	8193	Planet added via admin panel: 12-asdf	2025-12-02 00:54:16
30	1	DELETE_PLANET	planet	8193	Planet deleted from admin panel	2025-12-02 00:54:54

Planetpaper:

```
mysql> SELECT * FROM planetpaper;
```

planet_id	paper_id
32	4
34	5
34	6
29	8

4 rows in set (0.00 sec)

Observationfacility:

```
mysql> select * from observationfacility;
```

facility_id	name
32	Acton Sky Portal Observatory
17	Anglo-Australian Telescope
41	Apache Point Observatory
55	Arecibo Observatory
42	Atacama Large Millimeter Array (ALMA)
18	Bohunsan Optical Astronomical Observatory
24	Calar Alto Observatory
10	Cerro Tololo Inter-American Observatory
38	CHARacterising ExOPlanets Satellite (CHEOPS)
26	CoRoT
39	European Southern Observatory
12	European Space Agency (ESA) Gaia Satellite
40	Fred Lawrence Whipple Observatory
7	Gemini Observatory
68	Haleakala Observatory
34	HATNet
35	HATSouth
15	Haute-Provence Observatory
13	Hubble Space Telescope
59	Infrared Survey Facility
66	James Webb Space Telescope (JWST)
21	K2
43	KELT
45	KELT-North
44	KELT-South
14	Kepler
36	Kitt Peak National Observatory
46	KMTNet
47	KOINet
23	La Silla Observatory
49	Large Binocular Telescope Observatory
37	Las Campanas Observatory
58	Leoncito Astronomical Complex
6	Lick Observatory
70	Lowell Observatory
11	Mauna Kea Observatory
16	McDonald Observatory
30	MEarth Project
51	MOA
27	Multiple Facilities
5	Multiple Observatories
25	NASA Infrared Telescope Facility (IRTF)
52	Next-Generation Transit Survey (NGTS)
54	OGLE
3	Okayama Astrophysical Observatory
33	Palomar Observatory
9	Paranal Observatory
56	Parkes Observatory
57	Qatar
20	Roque de los Muchachos Observatory
50	South African Radio Astronomy Observatory (SAR)
48	SPECULOOS Southern Observatory
29	Spitzer Space Telescope
8	Subaru Telescope
62	SuperWASP
64	SuperWASP-North
63	SuperWASP-South
61	Teide Observatory
2	Thueringer Landessternwarte Tautenburg
19	Transiting Exoplanet Survey Satellite (TESS)
60	TrES
67	United Kingdom Infrared Telescope
72	University of Canterbury Mt John Observatory
31	Very Long Baseline Array
4	W. M. Keck Observatory
65	WASP-South
22	Wide-field Infrared Survey Explorer (WISE) Sat
53	Winer Observatory
1	Xinglong Station
69	XO
28	Yunnan Astronomical Observatory
71	Zwicky Transient Facility

Star

```
mysql> SELECT * FROM star;
```

star_id	system_id	name	spectral_type	temperature_k	radius_solar	mass_solar	metallicity_dex	surface_gravity	right_ascension_degree	declination_degree	distance_parsec	visual_magnitude
1	1	11 Com	G8 III	4874	13.76	2.89	-0.26	2.45	185.1787793	17.7932516	93.1846	4.72387
2	2	11 UMi	M4 III	4213	29.79	2.78	-0.82	1.93	229.2745954	71.8239428	125.321	5.813
3	3	14 And	M8 III	4888	11.55	1.78	-0.21	2.55	352.82415	39.2358367	75.4392	5.23133
4	4	14 Her	M8V	5338	0.93	0.91	0.4852	4.45	242.6021014	43.8163621	17.9323	6.61935
5	5	16 Cyg B	G3 V	5750	1.13	1.88	0.86	4.36	295.4656422	58.5168243	21.1397	6.215
6	6	17 Sco	M3 III	4157	25.92	1.22	-0.81	1.7	243.4628874	-11.8377985	124.953	5.22686
7	7	18 Del	G6 III	4980	8.8	2.1	-0.86	2.87	314.6078375	18.8391378	76.222	5.51048
8	8	1RXS J160929.1-210524	M7 V	4860	1.31316	0.85	NULL	3.99592	242.3762343	-21.0831483	139.135	12.618
9	9	24 Boo	G3 IV	4816	12.19	1.85	-0.78	2.28	217.1555242	49.8446488	95.9863	5.59
10	10	24 Sex	M8 IV	5098	4.9	1.54	-0.83	3.5	155.8684866	-8.9823998	72.0691	6.4535
11	11	2MASS J04372171+2651014	M V	3180	0.84	0.17	0.81	4.5	69.3485531	26.8562997	128.484	16.186
12	12	2MASS J01033563-5515561 A	M5.5/M6	NULL	NULL	0.19	NULL	NULL	15.8992489	-55.2658291	47.2	15.788
13	13	2MASS J01225093-2439505	M3.5 V	3538	0.36645	0.4	NULL	4.85884	20.7128012	-24.6646128	33.8281	14.244
14	14	2MASS J02192210-3925225	M6	3864	0.281	0.188	NULL	4.59	34.8427846	-39.4238767	40.88964117	NULL
15	15	2MASS J0249-8557 A	M6 V	2859	0.315959	0.046	NULL	4.98849	42.4851627	-5.9509921	66.0613	18.3

Planet:

```
mysql> SELECT * FROM planet;
```

planet_id	star_id	name	mass_earth	radius_earth	equilibrium_temp_k	disc_year	discovery_method	disc_facility	is_controversial	insolation_flux	image_path
1	1	11 Com b	0.238	0.914	890486	2007	Radial Velocity	Kinglong Station	0	NULL	assets/img/planets/cloud_giant.png
2	2	11 UMi b	0.88	4684.8142	NULL	2009	Radial Velocity	Thüringer Landessternwarte Tautenburg	0	NULL	assets/img/planets/cloud_giant.png
3	3	14 And b	0	1131.151381	NULL	2008	Radial Velocity	Okayama Astrophysical Observatory	0	NULL	assets/img/planets/cloud_giant.png
4	4	14 Her b	0.373	2559.472162	NULL	2002	Radial Velocity	W. M. Weck Observatory	0	NULL	assets/img/planets/cloud_giant.png
5	5	16 Cyg B b	0.40	565.7074	NULL	1996	Radial Velocity	Multiple Observatories	0	NULL	assets/img/planets/cloud_giant.png
6	6	17 Sco b	0.06	1373.018718	NULL	2020	Radial Velocity	Lick Observatory	0	NULL	assets/img/planets/cloud_giant.png
7	7	18 Del b	0.024	2926.246143	NULL	2008	Radial Velocity	Okayama Astrophysical Observatory	0	NULL	assets/img/planets/cloud_giant.png
8	8	1815 J160929.1-210824 b	NULL	3800	1700	2008	Imaging	Gemini Observatory	0	NULL	assets/img/planets/cloud_giant.png
9	9	24 Boo b	0.032	288.6424834	NULL	2018	Radial Velocity	Okayama Astrophysical Observatory	0	NULL	assets/img/planets/cloud_giant.png
10	10	24 Sex b	0.09	632.46	NULL	2010	Radial Velocity	Lick Observatory	0	NULL	assets/img/planets/cloud_giant.png
11	10	24 Sex c	0.29	273.32	NULL	2010	Radial Velocity	Lick Observatory	0	NULL	assets/img/planets/cloud_giant.png
12	11	2M0437 b	NULL	1271.313620	1450	2021	Imaging	Subaru Telescope	0	NULL	assets/img/planets/cloud_giant.png
13	12	2MASS J01033563-0515561 AB b	NULL	4111.79	NULL	2013	Imaging	Paranal Observatory	0	NULL	assets/img/planets/cloud_giant.png
14	13	2MASS J01225093-2439505 b	NULL	7706.5	1600	2013	Imaging	W. M. Weck Observatory	0	NULL	assets/img/planets/cloud_giant.png
15	14	2MASS J02192210-3926225 b	NULL	4417.837	NULL	2015	Imaging	Cerro Tololo Inter-American Observatory	0	NULL	assets/img/planets/cloud_giant.png
16	15	2MASS J0249-0557 c	NULL	3686.889521	NULL	2018	Imaging	Mauna Kea Observatory	0	NULL	assets/img/planets/cloud_giant.png
17	16	2MASS J03590986+2009361 b	NULL	9720.911326	2369	2024	Imaging	European Space Agency (ESA) Gaia Satellite	0	NULL	assets/img/planets/lava_world.png
18	17	2MASS J04414489+2301513 b	NULL	2383.6	NULL	2010	Imaging	Hubble Space Telescope	0	NULL	assets/img/planets/cloud_giant.png
19	18	2MASS J11011926-7732383 b	NULL	8009.195396	2677	2024	Imaging	European Space Agency (ESA) Gaia Satellite	0	NULL	assets/img/planets/lava_world.png
20	19	2MASS J11550405-7919100 b	NULL	6356.56014	2506	2024	Imaging	European Space Agency (ESA) Gaia Satellite	0	NULL	assets/img/planets/lava_world.png
21	20	2MASS J12073346-5932539 b	NULL	1589.15	1250	2004	Imaging	Paranal Observatory	0	NULL	assets/img/planets/cloud_giant.png
22	21	2MASS J19383200+4603591 b	0.33	603.877	NULL	2015	Eclipse Timing Variations	Kepler	0	NULL	assets/img/planets/cloud_giant.png
23	22	2MASS J21252752-0138278 b	NULL	3813.948804	1616	2024	Imaging	European Space Agency (ESA) Gaia Satellite	0	NULL	assets/img/planets/cloud_giant.png
24	23	2MASS J22362452+4751425 b	NULL	3972.875	NULL	2016	Imaging	W. M. Weck Observatory	0	NULL	assets/img/planets/cloud_giant.png
25	24	30 Ari B b	0.29	4392.4106	NULL	2009	Radial Velocity	Thüringer Landessternwarte Tautenburg	0	NULL	assets/img/planets/cloud_giant.png
26	25	4 UMa b	0.453	2511.400772	NULL	2006	Radial Velocity	Thüringer Landessternwarte Tautenburg	0	NULL	assets/img/planets/cloud_giant.png
27	26	47 UMa b	0.032	804.00	NULL	1996	Radial Velocity	Lick Observatory	0	NULL	assets/img/planets/cloud_giant.png
28	26	47 UMa c	0.090	171.621	NULL	2001	Radial Velocity	Lick Observatory	0	NULL	assets/img/planets/cloud_giant.png
29	26	47 UMa d	0.16	521.22	NULL	2009	Radial Velocity	Lick Observatory	0	NULL	assets/img/planets/cloud_giant.png
30	27	51 Eri b	0.57	3464.329636	700	2015	Imaging	Gemini Observatory	0	NULL	assets/img/planets/cloud_giant.png
31	28	51 Peg b	0.013	146.2018	NULL	1995	Radial Velocity	Aixte-Provence Observatory	0	NULL	assets/img/planets/cloud_giant.png
32	29	55 Cnc B b	0	3.5	NULL	2025	Radial Velocity	Mauna Kea Observatory	0	NULL	assets/img/planets/unknown.png
33	29	55 Cnc B c	0	5.3	NULL	2025	Radial Velocity	Mauna Kea Observatory	0	NULL	assets/img/planets/mini_neptune.png
34	30	55 Cnc b	0.0029	263.9705	700	1996	Radial Velocity	Lick Observatory	0	NULL	assets/img/planets/cloud_giant.png
35	30	55 Cnc c	0.008	54.4730	NULL	2004	Radial Velocity	McDonald Observatory	0	NULL	assets/img/planets/cloud_giant.png

Paper:

```
mysql> select paper_id, title, authors, year, abstract, uploaded_by, uploaded_at, file_name, length(file_data) as file_size_bytes from paper;
```

paper_id	title	authors	year	abstract	uploaded_by	uploaded_at	file_name	file_size_bytes
4	Characterizing planetary systems with SPIRou: questions about the magnetic cycle of 55 Cnc A and two new planets around B?	C. Moutou ¹ , P. Petit ¹ , P. Charpentier ¹ , P. Cristofari ² , C. Baruteau ¹ , P. Thébault ³ , L. Arnold ⁴ , E. Artigau ⁵ , A. Carmona ⁶ , I. N.J. Cook ⁵ , F. Debras ¹ , X. Delfosse ⁶ , J.-F. Donati ¹ , L. Malo ⁵ , and M. Ould-Elhadj ¹	2025		1	2025-11-19 13:36:07	55cnc.pdf	3898805
5	JWST reveals the rapid and strong day-side variability of 55 Cancri e	J. A. Patel ^{1,2} , A. Brandeker ¹ , D. Kitzmann ^{2,3}	2024		1	2025-11-19 13:37:40	55cnc2.pdf	5206373
6	A secondary atmosphere on the rocky exoplanet 55 Cancri e	Renyu Hu ^{1,2} , Aaron Bello-Arufe ¹	2024		1	2025-11-19 13:42:59	55cnc3.pdf	5355713
8	Retrograde-rotating Exoplanets Experience Obliquity Excitations in an Eccentricity-enabled Resonance	Steven M. Kreyche ¹ , Jason W. Barnes ¹	2021		1	2025-12-02 14:08:14	Kreyche_2020_Planet_Sci_J_1_8.pdf	2235226

4 rows in set (0.01 sec)

Orbit:

```
mysql> SELECT * FROM orbit;
```

planet_id	orbital_period_days	semimajor_axis	eccentricity
1	323.21	1.178	12.2
2	516.21997	1.53	12.3
3	186.76	0.775	13.1
4	1765.0389	2.774	12.6
5	798.5	1.66	13.5
6	578.38	1.45	12.9
7	982.85	2.476	12.5
9	30.33	0.194	13.9
10	452.8	1.333	13.4
11	883	2.08	13.9
22	406	0.92	13.4
25	335.10001	0.99	12.3
26	270.27	0.877	12.6
27	1078	2.1	13.2
28	2391	3.6	14.2
29	14002	11.6	13.5
30	9100	10.4	NULL
31	4.230785	0.0527	14.3
32	6.7994	0.044	1.69
33	33.747	0.13	2.16
34	14.651552	0.118	13.9
35	44.3936	0.247	8.51
36	4799	5.6	13
37	0.7365474	0.01544	1.875
38	260.58	0.802	7.59
39	919.86	2.028	13.4
40	4.215	0.050201	2.11

Planetary_system

```
mysql> SELECT * FROM planetary_system;
```

system_id	system_name	num_of_stars	num_of_planets
1	11 Com	2	1
2	11 UMi	1	1
3	14 And	1	1
4	14 Her	1	2
5	16 Cyg B	3	1
6	17 Sco	1	1
7	18 Del	2	1
8	1RXS J160929.1-210524	1	1
9	24 Boo	1	1
10	24 Sex	1	2
11	2MASS J04372171+2651014	1	1
12	2MASS J01033563-5515561 A	2	1
13	2MASS J01225093-2439505	1	1
14	2MASS J02192210-3925225	1	1
15	2MASS J0249-0557 A	2	1
16	2MASS J03590986+2009361	1	1
17	2MASS J04414489+2301513	1	1
18	2MASS J11011926-7732383	1	1
19	2MASS J11550485-7919108	1	1
20	2MASS J12073346-3932539	1	1
21	2MASS J19383260+4603591	2	3
22	2MASS J21252752-8138278	1	1
23	2MASS J22362452+4751425	1	1
24	30 Ari B	4	1
25	4 UMa	1	1
26	47 UMa	1	3
27	51 Eri	3	1
28	51 Peg	1	1
29	55 Cnc B	2	7
30	55 Cnc	2	7
31	6 Lyn	1	1
32	61 Vir	1	3
33	7 CMa	1	2
34	70 Vir	1	1
35	75 Cet	2	2
36	8 UMi	1	1
37	81 Cet	1	1
38	91 Aqr	3	1
39	AB Aur	1	1
40	AB Pic	1	1

Pattern Identification Queries:

1) Advanced planet search with multiple filters

```
SELECT p.planet_id, p.name AS planet, s.name AS host_star, d.disc_year,
dm.method_name,

f.name AS facility_name, s.distance_parsec,

o.orbital_period_days, p.mass_earth, p.radius_earth

FROM planet AS p

LEFT JOIN star AS s ON s.star_id= p.star_id

LEFT JOIN discovery AS d ON d.planet_id = p.planet_id

LEFT JOIN discoverymethod AS dm ON dm.method_id = d.method_id

LEFT JOIN observationfacility AS f ON f.facility_id = d.facility_id

LEFT JOIN orbit AS o ON o.planet_id = p.planet_id

WHERE (p.name LIKE CONCAT('%', ?, '%')

      OR s.name LIKE CONCAT('%', ?, '%'))

AND (d.disc_year >= ? OR ? IS NULL)

AND (d.disc_year <= ? OR ? IS NULL)

AND (dm.method_name LIKE CONCAT('%', ?, '%') OR ? IS NULL)

AND (f.name LIKE CONCAT('%', ?, '%') OR ? IS NULL)

ORDER BY d.disc_year DESC,

       p.name ASC

LIMIT ? OFFSET ?;
```

2) Hottest 10 planets with star context

```
SELECT p.planet_id, p.name AS planet, s.name AS host_star,
       p.equilibrium_temp_k AS planet_temp_k,
       s.temperature_k AS star_temp_k,
       s.distance_parsec,
       d.disc_year,
       dm.method_name
FROM planet AS p
JOIN star AS s ON s.star_id = p.star_id
LEFT JOIN discovery AS d ON d.planet_id = p.planet_id
LEFT JOIN discoverymethod AS dm ON dm.method_id = d.method_id
WHERE p.equilibrium_temp_k IS NOT NULL
ORDER BY p.equilibrium_temp_k DESC,
         s.temperature_k DESC
LIMIT 10;
```

3) Most packed star systems

```
SELECT ps.system_id, ps.system_name,  
       COUNT(DISTINCT s.star_id) AS star_count,  
       COUNT(DISTINCT p.planet_id) AS planet_count  
FROM planetary_system AS ps  
LEFT JOIN star AS s ON s.system_id = ps.system_id  
LEFT JOIN planet AS p ON p.star_id = s.star_id  
GROUP BY ps.system_id, ps.system_name  
HAVING planet_count >= 1  
ORDER BY planet_count DESC, star_count DESC, ps.system_name ASC  
LIMIT 10;
```

4) Top 5 most packed individual stars

```
SELECT s.star_id, s.name AS star_name, ps.system_name,  
       COUNT(p.planet_id) AS planet_count,  
       AVG(p.mass_earth) AS avg_mass_earth,  
       AVG(p.radius_earth) AS avg_radius_earth  
FROM star AS s  
LEFT JOIN planetary_system AS ps ON ps.system_id = s.system_id  
LEFT JOIN planet AS p ON p.star_id = s.star_id  
GROUP BY s.star_id, s.name, ps.system_name  
HAVING planet_count >= 1  
ORDER BY planet_count DESC, avg_mass_earth DESC,  
         s.name ASC  
LIMIT 5;
```

5) Heaviest planetary systems

```
SELECT ps.system_id, ps.system_name,  
       COUNT(DISTINCT p.planet_id) AS planet_count,  
       SUM(p.mass_earth) AS total_mass_earth,  
       AVG(p.mass_earth) AS avg_mass_earth  
FROM planetary_system AS ps  
JOIN star AS s ON s.system_id = ps.system_id  
JOIN planet AS p ON p.star_id = s.star_id  
WHERE p.mass_earth IS NOT NULL  
GROUP BY ps.system_id, ps.system_name  
HAVING planet_count >= 2  
ORDER BY total_mass_earth DESC, avg_mass_earth DESC;
```

6) Discovery trend by year and method

```
SELECT d.disc_year, dm.method_name,  
       COUNT(DISTINCT d.planet_id) AS planets_discovered,  
       COUNT(d.disc_id) AS discovery_records  
FROM discovery AS d  
JOIN discoverymethod AS dm ON dm.method_id = d.method_id  
GROUP BY d.disc_year, dm.method_name  
HAVING d.disc_year IS NOT NULL  
ORDER BY d.disc_year ASC,  
         planets_discovered DESC,  
         dm.method_name ASC;
```

7) Planets with number of linked research papers

```
SELECT p.planet_id, p.name AS planet, s.name AS host_star,
       COUNT(pp.paper_id) AS paper_count,
       MIN(pa.year) AS earliest_paper_year,
       MAX(pa.year) AS latest_paper_year
FROM planet AS p
JOIN star AS s ON s.star_id = p.star_id
LEFT JOIN planetpaper AS pp ON pp.planet_id = p.planet_id
LEFT JOIN paper AS pa ON pa.paper_id = pp.paper_id
GROUP BY p.planet_id, p.name, s.name
ORDER BY paper_count DESC, latest_paper_year DESC, p.name ASC;
```

8) Recent paper uploads per day (last 10 days)

```
SELECT DATE(pa.uploaded_at) AS upload_day,
       COUNT(pa.paper_id) AS paper_count,
       COUNT(DISTINCT pa.uploaded_by) AS distinct_uploaders
FROM paper AS pa
WHERE pa.uploaded_at >= DATE_SUB(CURDATE(), INTERVAL 10 DAY)
GROUP BY DATE(pa.uploaded_at)
ORDER BY upload_day DESC;
```

9)Admin activity summary from audit log

```
SELECT u.username, u.role,
       COUNT(l.log_id) AS total_actions,
       SUM(CASE WHEN l.action = 'LOGIN' THEN 1 ELSE 0 END) AS login_count,
       SUM(CASE WHEN l.action = 'ADD_PAPER' THEN 1 ELSE 0 END) AS papers_added,
       SUM(CASE WHEN l.action LIKE 'DELETE_%' THEN 1 ELSE 0 END) AS delete_ops
FROM auditlog AS l
JOIN appuser AS u ON u.user_id = l.user_id
WHERE l.created_at >= DATE_SUB(CURDATE(), INTERVAL 7 DAY)
GROUP BY u.username, u.role
ORDER BY
       total_actions DESC,
       u.username ASC;
```

10) User login statistics (logins per user)

```
SELECT u.user_id, u.username, u.role,
       COUNT(l.log_id) AS login_count,
       MIN(l.created_at) AS first_login,
       MAX(l.created_at) AS last_login
FROM appuser AS u
LEFT JOIN auditlog AS l ON l.user_id = u.user_id AND l.action = 'LOGIN'
GROUP BY u.user_id, u.username, u.role
ORDER BY login_count DESC, last_login DESC;
```

11) Potentially habitable-zone planets

```
SELECT p.planet_id, p.name AS planet, s.name AS host_star, p.insolation_flux,  
       p.equilibrium_temp_k, s.spectral_type, s.distance_parsec  
FROM planet AS p  
JOIN star AS s ON s.star_id = p.star_id  
WHERE p.insolation_flux BETWEEN 0.5 AND 1.5  
      AND p.equilibrium_temp_k BETWEEN 240 AND 320  
ORDER BY p.insolation_flux ASC, p.equilibrium_temp_k ASC;
```

12) Observation statistics per facility

```
SELECT f.facility_id,  
       f.name AS facility_name,  
       COUNT(o.obs_id) AS obs_count,  
       AVG(o.exposure_time_s) AS avg_exposure_s,  
       AVG(o.snr) AS avg_snr,  
       MIN(o.obs_date) AS first_obs_date,  
       MAX(o.obs_date) AS last_obs_date  
FROM observationfacility AS f  
LEFT JOIN observation AS o ON o.facility_id = f.facility_id  
GROUP BY f.facility_id, f.name  
ORDER BY obs_count DESC, avg_snr DESC,  
         facility_name ASC;
```

13) Planets with discovery

```
SELECT p.planet_id, p.name AS planet, s.name AS host_star, d.disc_year,  
       dm.method_name, f.name AS facility_name  
FROM planet AS p  
JOIN star AS s  ON s.star_id  = p.star_id  
JOIN discovery AS d  ON d.planet_id = p.planet_id  
LEFT JOIN orbit AS o  ON o.planet_id = p.planet_id  
LEFT JOIN discoverymethod AS dm  ON dm.method_id = d.method_id  
LEFT JOIN observationfacility AS f  ON f.facility_id = d.facility_id  
WHERE o.planet_id IS NULL  
ORDER BY d.disc_year DESC, p.name ASC;
```

14) Systems ranked by total observational effort

```
SELECT ps.system_id, ps.system_name,  
       COUNT(DISTINCT o.obs_id) AS total_observations,  
       COUNT(DISTINCT p.planet_id) AS observed_planets,  
       SUM(o.exposure_time_s), AS total_exposure_s,  
       AVG(o.snr) AS avg_snr  
FROM planetary_system AS ps  
JOIN star AS s ON s.system_id = ps.system_id  
JOIN planet AS p ON p.star_id = s.star_id  
JOIN observation AS o ON o.planet_id = p.planet_id  
GROUP BY ps.system_id, ps.system_name  
HAVING total_observations >= 1  
ORDER BY total_observations DESC, total_exposure_s DESC;
```

15) Recently discovered planets that already have papers

```
SELECT p.planet_id, p.name AS planet, s.name AS host_star, d.disc_year,
       COUNT(DISTINCT pa.paper_id) AS paper_count,
       MIN(pa.year) AS first_paper_year
FROM planet AS p
JOIN star AS s ON s.star_id = p.star_id
JOIN discovery AS d ON d.planet_id = p.planet_id
JOIN planetpaper AS pp ON pp.planet_id = p.planet_id
JOIN paper AS pa ON pa.paper_id = pp.paper_id
WHERE d.disc_year >= 2015
GROUP BY p.planet_id, p.name, s.name, d.disc_year
ORDER BY d.disc_year DESC, paper_count DESC;
```

16) Planets observed by multiple facilities

```
SELECT p.planet_id, p.name AS planet, s.name AS host_star,
       COUNT(DISTINCT o.facility_id) AS facility_count,
       COUNT(o.obs_id) AS total_observations
FROM planet AS p
JOIN star AS s ON s.star_id = p.star_id
JOIN observation AS o ON o.planet_id = p.planet_id
GROUP BY p.planet_id, p.name, s.name
HAVING facility_count >= 2
ORDER BY facility_count DESC, total_observations DESC,
       p.name ASC;
```

17) Planets that are heavier than average for their star

```
SELECT p.planet_id, p.name AS planet, s.name AS host_star,
       p.mass_earth, star_stats.avg_mass_per_star
FROM planet AS p JOIN star AS s ON s.star_id = p.star_id
JOIN (SELECT p2.star_id, AVG(p2.mass_earth) AS avg_mass_per_star
      FROM planet AS p2
      WHERE p2.mass_earth IS NOT NULL
      GROUP BY p2.star_id
) AS star_stats ON star_stats.star_id = p.star_id
WHERE p.mass_earth IS NOT NULL
      AND p.mass_earth > star_stats.avg_mass_per_star
ORDER BY (p.mass_earth - star_stats.avg_mass_per_star) DESC,
         s.name ASC;
```

18) NASA Dataset import:

LOAD DATA INFILE 'C:\ProgramData\MySQL\MySQL Server
8.0\Uploads/clean_exoplanets.csv'

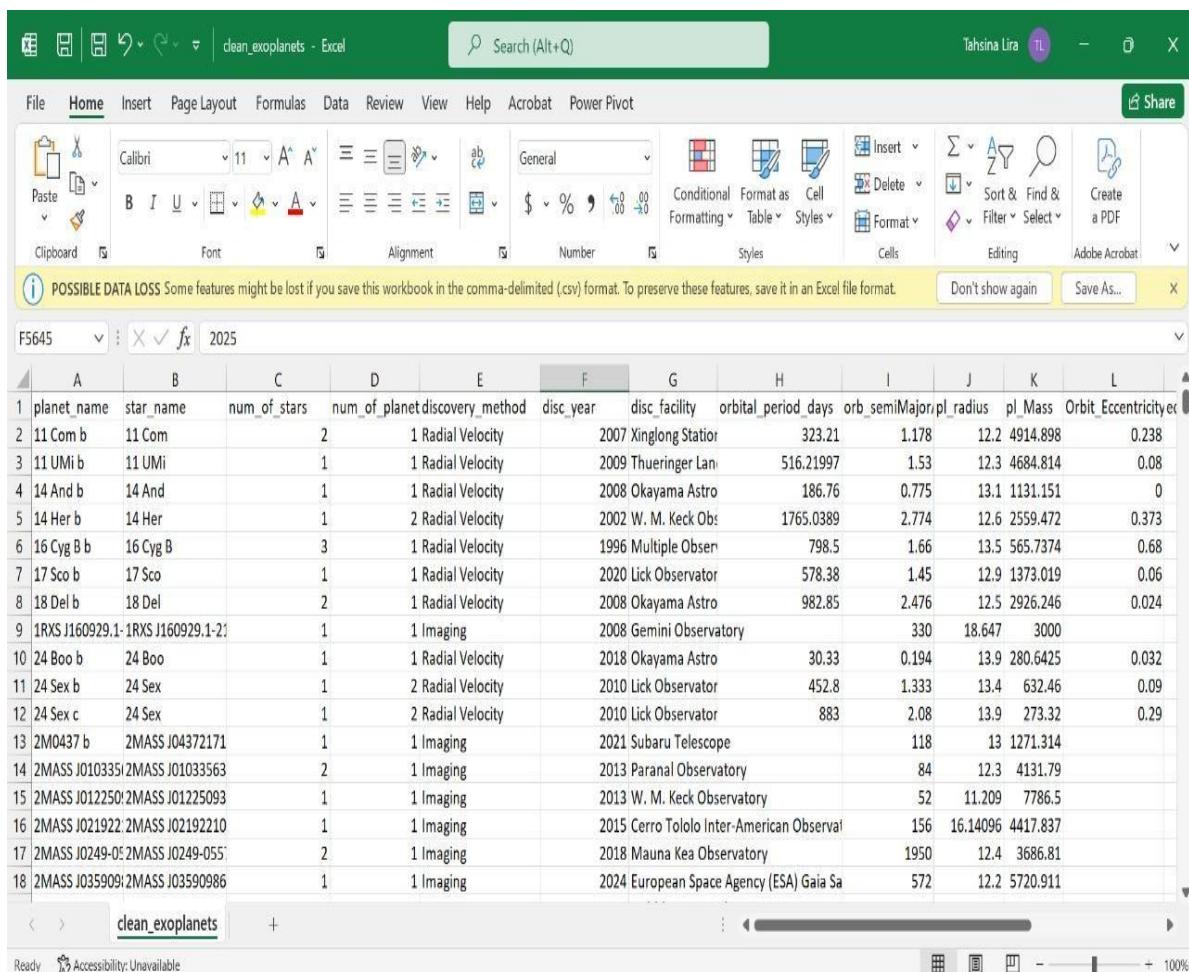
INTO TABLE planet

FIELDS TERMINATED BY ','

ENCLOSED BY ''

LINES TERMINATED BY '\n'

IGNORE 1 ROWS;



	A	B	C	D	E	F	G	H	I	J	K	L
	planet_name	star_name	num_of_stars	num_of_planet	discovery_method	disc_year	disc_facility	orbital_period_days	orb_semiMajor	pl_radius	pl_Mass	Orbit_Eccentricity
2	11 Com b	11 Com	2	1	Radial Velocity	2007	Xinglong Station	323.21	1.178	12.2	4914.898	0.238
3	11 UMi b	11 UMi	1	1	Radial Velocity	2009	Thuringer Land	516.21997	1.53	12.3	4684.814	0.08
4	14 And b	14 And	1	1	Radial Velocity	2008	Okayama Astro	186.76	0.775	13.1	1131.151	0
5	14 Her b	14 Her	1	2	Radial Velocity	2002	W. M. Keck Obs	1765.0389	2.774	12.6	2559.472	0.373
6	16 Cyg B b	16 Cyg B	3	1	Radial Velocity	1996	Multiple Obser	798.5	1.66	13.5	565.7374	0.68
7	17 Sco b	17 Sco	1	1	Radial Velocity	2020	Lick Observatory	578.38	1.45	12.9	1373.019	0.06
8	18 Del b	18 Del	2	1	Radial Velocity	2008	Okayama Astro	982.85	2.476	12.5	2926.246	0.024
9	1RXS J160929.1-1	1RXS J160929.1-2	1	1	Imaging	2008	Gemini Observatory		330	18.647	3000	
10	24 Boo b	24 Boo	1	1	Radial Velocity	2018	Okayama Astro	30.33	0.194	13.9	280.6425	0.032
11	24 Sex b	24 Sex	1	2	Radial Velocity	2010	Lick Observatory	452.8	1.333	13.4	632.46	0.09
12	24 Sex c	24 Sex	1	2	Radial Velocity	2010	Lick Observatory	883	2.08	13.9	273.32	0.29
13	2M0437 b	2MASS J04372171	1	1	Imaging	2021	Subaru Telescope		118	13	1271.314	
14	2MASS J010335	2MASS J01033563	2	1	Imaging	2013	Paranal Observatory		84	12.3	4131.79	
15	2MASS J012250	2MASS J01225093	1	1	Imaging	2013	W. M. Keck Observatory		52	11.209	7786.5	
16	2MASS J021922	2MASS J02192210	1	1	Imaging	2015	Cerro Tololo Inter-American Observa		156	16.14096	4417.837	
17	2MASS J0249-0	2MASS J0249-055	2	1	Imaging	2018	Mauna Kea Observatory		1950	12.4	3686.81	
18	2MASS J035909	2MASS J03590986	1	1	Imaging	2024	European Space Agency (ESA) Gaia Sa		572	12.2	5720.911	

Progress Report

Summary:

- ☐ • Successfully designed a complete relational database for the ExoDiscover system.
- ☐ • Implemented normalization for all tables up to **Third Normal Form (3NF)**.
- ☐ • Developed a fully functioning **web application** using **Java Servlets and JSP**.
- ☐ • Integrated the system with **MySQL**, including advanced SQL queries and complex JOIN operations.
- ☐ • Created interactive **dashboards** with analytical visualizations for planets, stars, and discovery trends.
- ☐ • Each group member actively contributed to database design, interface development, backend logic, and documentation.
- ☐ • Ensured smooth project development through **structured planning** and consistent weekly progress.
- ☐ • Maintained regular communication, enabling fast problem-solving and shared understanding of tasks.
- ☐ • Achieved equal participation from all members, leading to a balanced and well-coordinated project outcome.

Initial Planning:

Team Member	Total Hours (period)	Remarks
	05.10.2025 – 31.10.2025 (08 hours 30 mins)	Initial project planning, ER diagram brainstorming, basic schema ideas. Reviewed NASA exoplanet dataset and discussed project scope.
	05.10.2025 – 31.10.2025 (09 hours 00 mins)	Literature review on exoplanet catalogues, noted attributes for database. Helped to sketch early conceptual model and user roles (admin, user, guest).
Group Totals	17 hours 30 mins	Combined time spent on idea selection and early database design.

Individual Report – Member 1

Tahsina Hossain Lira

Reporting Period: 05.10.2025 – 04.12.2025

Date	Hours	Activities
05.10.2025	2hours	Read course notes on DBMS and revised normalization concepts.
8.10.2025	2 hours 30 min	Analysed NASA exoplanet CSV file; listed useful attributes for tables.
12.10.2025	3 hours	Drew rough ER diagram for planets, stars, systems, and discovery details.
16.10.2015	1 hour 45 min	Discussed ER changes with teammate; updated cardinalities and keys.
20.10.2025	2 hours 30 min	Discussed ER changes with teammate; updated cardinalities and keys.
24.10.2025	2 hours 15 min	Created tables in MySQL and tested basic INSERT and SELECT queries.
28.10.2025	2 hours 15 min	Imported sample rows from Excel into staging_exo table and cleaned data.
02.11.2025	2 hours 20 mins	Worked on joins between planet, star, and planetary_system tables.
06.11.2025	2 hours 40 mins	Helped design search filters (year, method, facility) for user dashboard.
10.11.2025	2 hours 30 mins	Wrote explanation of 1NF, 2NF, 3NF for core tables in the project report.

18.11.2025	3 hours	Tested complex queries for hottest planets and most packed star systems.
24.11.2025	2 hours 20 mins	Reviewed and corrected schema, checked foreign keys and constraints.
30.11.2025	2 hours 10 mins	Updated documentation (summary, goals, limitations, future work).
04.12.2025	2 hours	Final review of SQL scripts and database backup before submission.

Individual Report – Member 2

Israt Jahan

Reporting Period: 05.10.2025 – 04.12.2025

Date	Hours	Activities
06.10.2025	2 hours 15 mins	Studied JSP and servlet basics; set up Apache Tomcat test project.
09.10.2025	2 hours 30 mins	Implemented simple index.jsp welcome page for ExoDiscover.
13.10.2025	3 hours 00 mins	Created Register and Login forms with basic validation on the client side.
17.10.2025	2 hours 20 mins	Implemented LoginServlet and password hashing logic connected to appuser table.
21.10.2025	2 hours 45 mins	Built user dashboard layout and integrated welcome message using session data.

25.10.2025	2 hours 30 mins	Implemented search servlet using complex WHERE clauses and pagination.
29.10.2025	2 hours 40 mins	Added planet details page and linked research paper section.
03.11.2025	2 hours 30 mins	Developed admin dashboard with navigation to planets, stars, papers, and logs.
08.11.2025	2 hours 50 mins	Implemented paper upload servlet using @MultipartConfig and BLOB storage.
14.11.2025	2 hours 30 mins	Created audit logging utility and integrated it with login and paper actions.
19.11.2025	3 hours 00 mins	Designed logs view (admin_logs.jsp) with filters and summary side panels.
23.11.2025	2 hours 35 mins	Implemented Chart.js discovery trend graph on user dashboard.
28.11.2025	2 hours 15 mins	Helped polish CSS theme, starfield background, and responsive layout tweaks.
04.12.2025	2 hours 20 mins	Prepared presentation slides and tested full project workflow end-to-end.

Learning outcomes

Technologies and Skills Learned

(Report Version – for Final Project Submission)

During the development of the ExoDiscover – Exoplanet Exploration and Research Management System, a wide range of technical skills, database concepts, and software development tools were learned and applied. The project combines backend logic, relational database design, and interactive user interfaces to provide a complete scientific data exploration experience. The key technologies and skills gained are outlined below:

1. Java (Servlets, JSP, and OOP Concepts)

Java served as the core backend programming language for building the functional logic of ExoDiscover. Through this project, we gained strong proficiency in:

- **Object-Oriented Programming (OOP)**

Developing model classes such as Planet, Star, Paper, PlanetSummary, and AuditLog strengthened our skills in:

Class and object design

Encapsulation and abstraction

Inheritance and code reusability

Mapping real-world scientific entities into Java objects

- **Servlet-Based Web Development**

We implemented request handling, form submission, routing, and session validation using Java Servlets:

GET and POST request processing

Login authentication and session creation

Role-based access control for Admin, User, and Guest

Controller logic for dashboards and analytics

- **JSP & JSTL**

JavaServer Pages were used to build dynamic web interfaces:

Displaying database results

Implementing loops and conditions using JSTL tags

Creating reusable UI components such as headers and menus

- **Exception and Session Handling**

Robust error-handling techniques were used to ensure smooth operation and stable user sessions.

2. MySQL (Database Design, SQL Queries, and DBMS Concepts)

MySQL was used as the main relational database system, and its integration added significant learning outcomes.

- **Database Design & Normalization**

We designed a multi-table database including entities such as:

User, Star, Planet, Orbit, Observation, Discovery

Paper, PlanetPaper (junction table)

AuditLog, DiscoveryMethod, ObservationFacility

The entire schema was developed according to Third Normal Form (3NF) ensuring:

No partial dependencies

No transitive dependencies

Minimal redundancy

Strict referential integrity

- **Advanced SQL Querying**

The project involved writing and optimizing complex SQL queries, including:

Multi-table JOIN (4–5 table joins)

Aggregation (COUNT, AVG, MAX, GROUP BY)

Subqueries and filtering

Dynamic search with optional filters

View creation (e.g., v_planet_summary)

These queries power features like:

Discovery trends

Top hottest planets

Top packed star systems

Paper uploads per day

- **Foreign Keys & Constraints**

Work on relational integrity deepened understanding of:

ON DELETE CASCADE / RESTRICT

Primary key design

Composite keys in junction tables

Enforcing valid relationships between entities

3. Front-End Development (HTML, CSS & JavaScript)

- **HTML Structure & Page Layout**

We created a fully structured layout with multiple JSP-powered pages including:

Login, Register, Guest View

User Dashboard, Admin Dashboard

Planet Search, Star Systems, Logs, Paper Uploads

Data Visualization pages

- **CSS Styling & UI/UX**

We learned styling concepts including:

Theme design (galaxy/starfield UI)

Components like cards, tables, grids, and dashboards

Responsive page layouts using Flexbox and Grid

Custom UI design for modern, clean presentation

- **JavaScript Enhancements**

JavaScript was used for:

Live animated starfield background (space-bg.js)

Chart.js visualizations (discovery trends)

Interactive elements such as buttons and dynamic content updates

4. Chart.js & Data Visualization

Chart.js allowed the creation of scientific visualizations for dashboard analytics:

Line charts for discovery trends

Bar charts for star/planet statistics

Real-time data representation driven by SQL

This enriched the system's usability and analytical depth.

Limitations of This Project (ExoDiscover)

Scalability:

Database Performance:

As the volume of exoplanet, star, and research data grows, certain queries—especially multi-table joins—may slow down. Further optimization such as indexing, caching, and query tuning will be required for handling large scientific datasets.

Multi-User Access:

The current version is designed for local use. If many users (admins, researchers, guests) access the system at the same time, performance may degrade without a cloud-based or load-balanced architecture.

Security:

Basic Authentication Only:

The system uses simple username–password login with role-based access. It does not yet support advanced security such as two-factor authentication, HTTPS, rate limiting, or IP monitoring.

File Storage Risks:

Uploaded research papers (PDFs) are stored in the database as BLOBs. There is no file encryption or malware validation, which may pose security concerns in large-scale deployment.

User Experience:

Limited UI Modernization:

Although the interface is themed and user-friendly, it lacks advanced UI components such as interactive visualizations, animations, and responsive design for various screen sizes.

No Mobile Support:

The system works best on desktop browsers. Guest and admin dashboards are not optimized for mobile or tablet devices.

Features:**Restricted Real-time Integration:**

ExoDiscover does not fetch live data from NASA or other scientific APIs. All dataset imports are manual, which limits real-time authenticity.

Limited Scientific Analysis Tools:

Advanced computations—such as habitability prediction, orbital simulation, or AI-assisted classification—are not included due to scope constraints.

Observation Module Not Fully Implemented:

The observation table contains no real telescope records. Without sensor readings, many scientific insights cannot be visualized accurately.

Customization & Flexibility:**Fixed Model Structure:**

The system architecture is tailored to one dataset. Adding more observational metrics, more star catalogs (Kepler, TESS), or new discovery techniques would require structural changes.

Hard-coded Role System:

Admin and user roles are predefined. Adding more roles (e.g., researcher, reviewer) requires additional backend development.

Conclusion

ExoDiscover successfully demonstrates how a structured Database Management System can be used to organize, explore, and analyze scientific data related to exoplanets and star systems. By integrating MySQL with a Java-based web application, the system offers a unified platform for managing planets, stars, discovery methods, orbital details, and research papers. The addition of features such as dashboards, search tools, complex SQL queries, role-based access (admin/user/guest), and audit logs enhances both functionality and scientific usefulness.

Although the project works efficiently with clean data modeling and 3NF-compatible tables, certain advanced features—such as real-time NASA API integration, AI-based habitability prediction, and observation data visualization—remain outside the current scope. Despite these limitations, ExoDiscover lays a strong foundation for a scalable scientific information system and demonstrates practical understanding of relational databases, system design, backend development, and data visualization.

Overall, the project achieves its goal of creating an interactive exoplanet exploration system that is educational, research-friendly, and technically sound, while also offering clear opportunities for future enhancement through automation, cloud deployment, and advanced analytics.